

SIMPLE LINEAR REGRESSION

Motivation [READING]

In many cases it is necessary to make decisions, assess scenarios, define business strategies or political actions under conditions of uncertainty.

Such decisions are often based on the (hypothesised) relationship between a variable of interest and one or more variables supposedly linked to it.

- ▶ After assessing the relationship between sales and expenditure on advertising on TV, radio and social media, the marketing manager of a company can use this relationship to predict sales given certain levels of advertising expenditure in the different media.
- ▶ The relationship between electricity demand and daytime temperatures can be used to predict next month's electricity use from daytime temperature forecasts

Regression analysis

Regression analysis is a statistical procedure aimed at determining, on the basis of an appropriate set of (sample) data, an equation that describes – or, better, **estimates** - the relationship between variables, and specifically the relationship between

- ▶ a **dependent variable (or response variable)** - conventionally denoted Y - that one is interested in explaining/predicting as a function of
- ▶ one or more **independent (or explanatory) variables** - denoted by $X_1, X_2, \dots, X_p$

We start considering the **simple regression model**, i.e. the case when

- ▶ the dependent variable Y **must be explained/predicted on** the basis of a **single** explanatory variable, X .
- ▶ Y and X are both numerical

Simple linear regression model for the population [READING]

Example. A financial advisory firm has set up an online advisory service and, on an experimental basis, has started an online chat service for its clients, where they exchange ideas or information on investments. Given the success of the initiative, the company is considering selling advertising space (banners) to other companies.

Certainly, crucial aspects concern the number of active users, the possible evolution of the number of users over time, etc.

However, it is of interest to start analysing the characteristics of the users, and to study the relationship between the time spent on chat and some socio-demographic characteristics, in order to provide companies interested in purchasing advertising spaces with information on the characteristics of the users who use the chat the most, in order to assess whether or not the platform's users are on target.

The simple linear regression model for the population

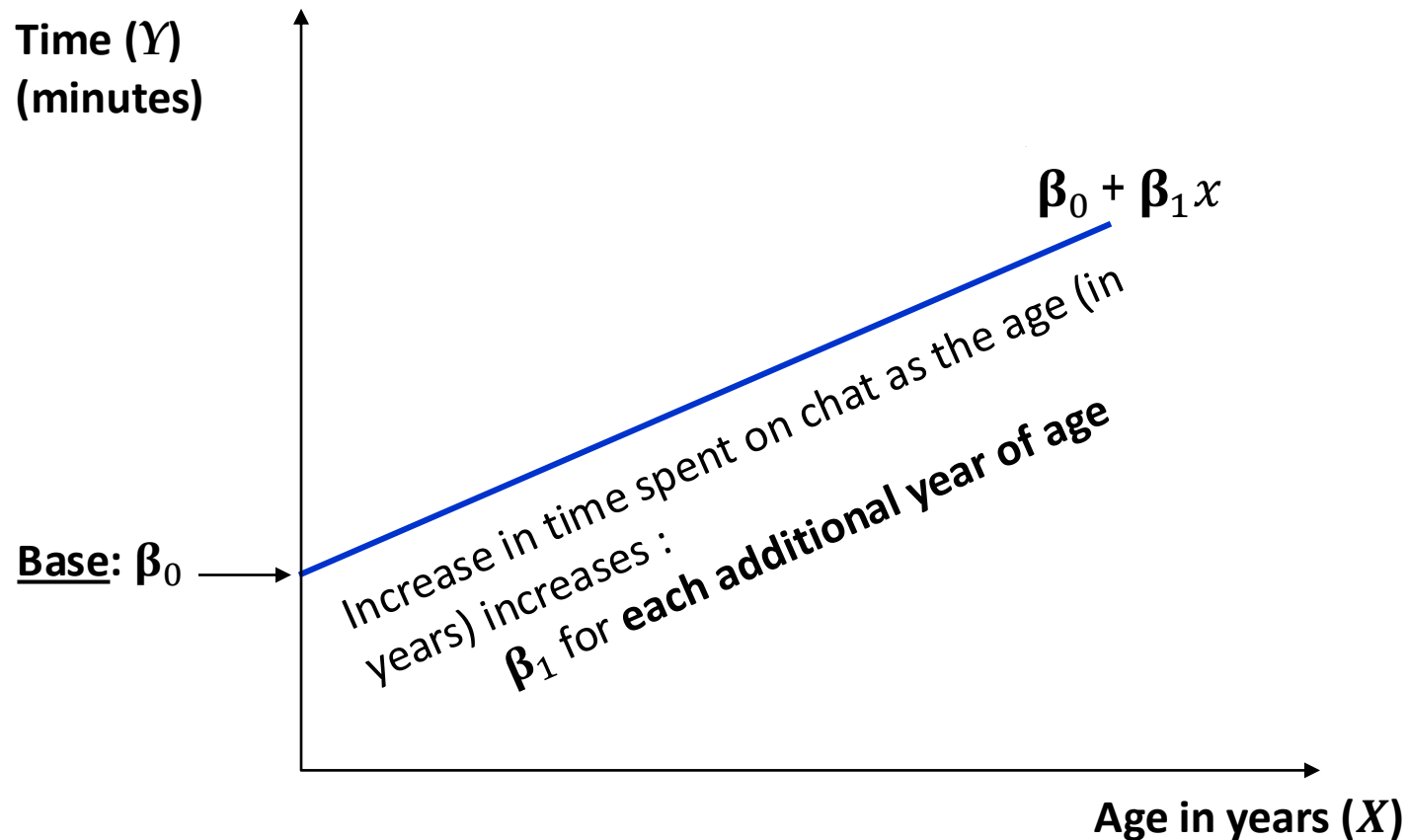
Given the target group of a specific set of companies, we are interested in studying the relationship between age and time spent chatting by focusing on users under 60 years of age. It is considered reasonable to **assume** that:

- **Time spent on chat** depends (also) on age
- The relationship between **time spent on chat** and **age** is direct.
- The relationship between **time spent on chat** and **age**, at least with reference to the focal age group, is linear.

These assumptions imply that **there is a linear model** linking **time spent on chat** to **age**

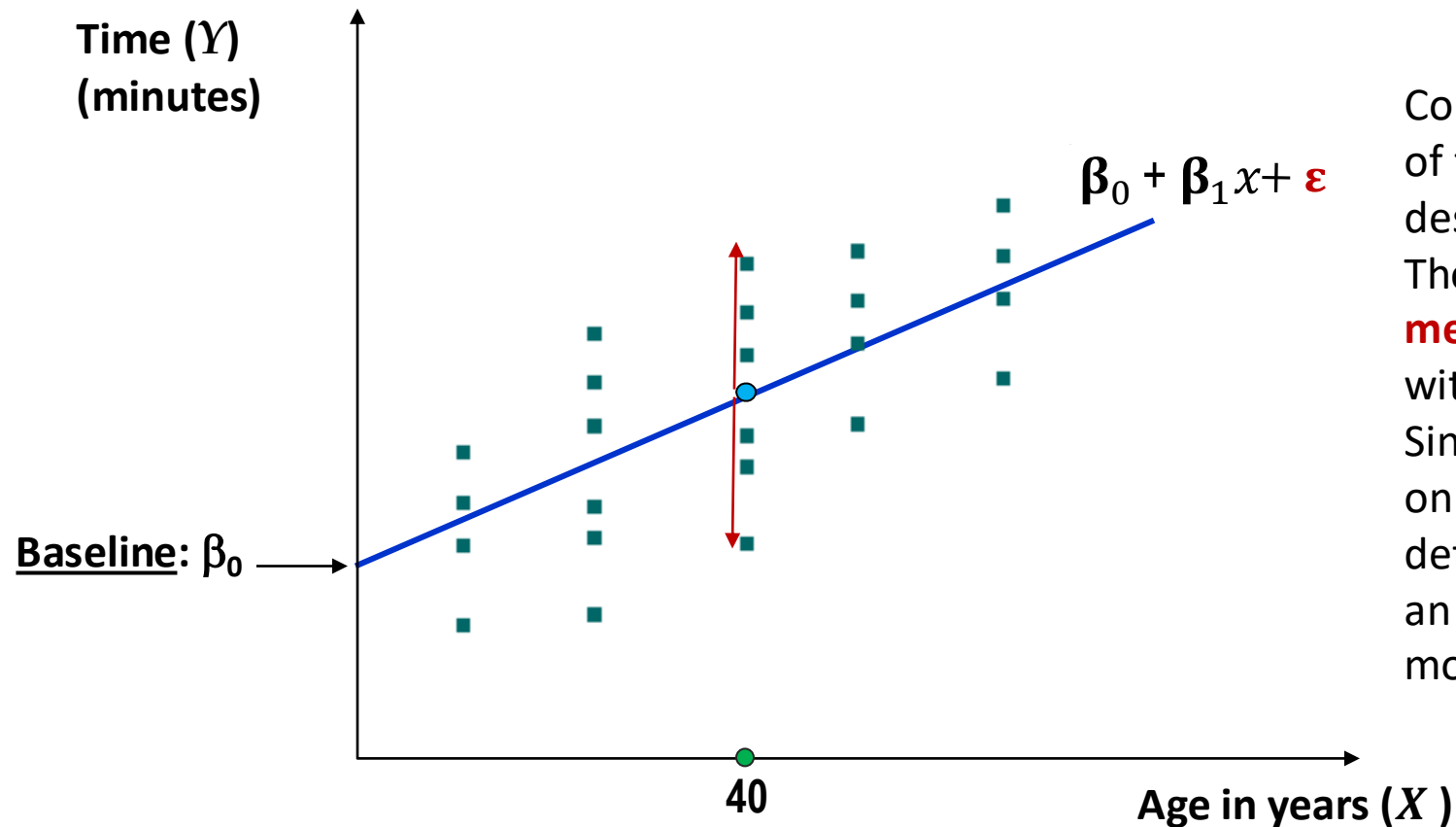
The simple linear regression model for the population

Linear model describing the relationship between the dependent variable and the values of the explanatory variable in the population (**deterministic** component)



The simple linear regression model for the population

A **probabilistic component** is added to the linear model, to describe the different values taken by the dependent variable corresponding to the same value of the explanatory variable



Consider users aged **40**: the time on chat of these users will be different (random, described by ϵ).

The linear model can only describe the **mean** (or expected value) of Y for users with different ages.

Since, given a certain age, the time spent on chat by different users **varies in a** non-deterministic and unpredictable manner, an **erratic** component is added to the model.

The simple linear regression model for the population

The simple linear regression model of a random variable Y given a specific value of the explanatory variable is defined as:

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

Y = dependent variable (*random*)

x = value taken by the independent variable (*deterministic, known*)

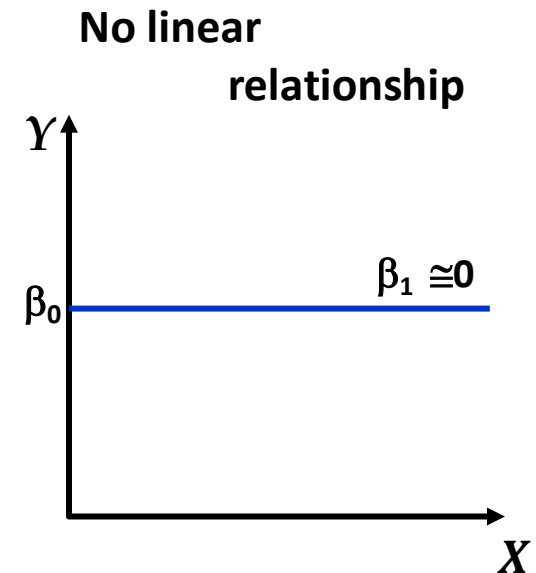
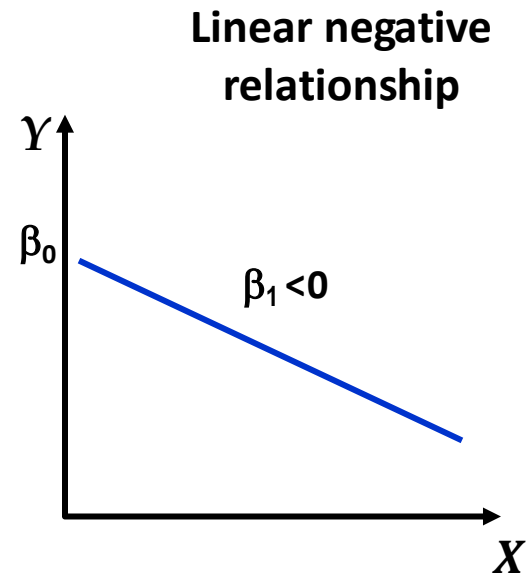
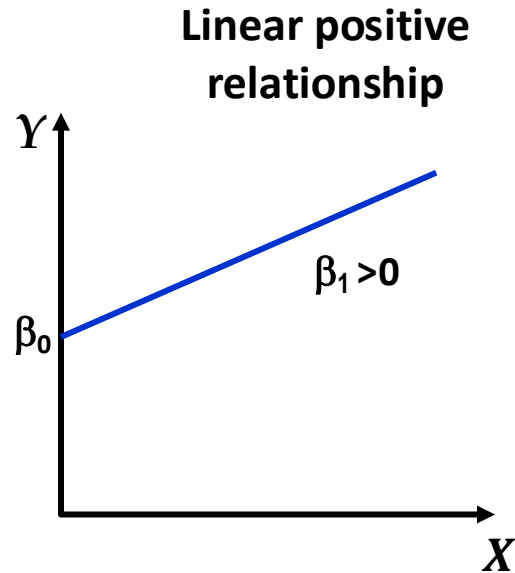
β_0, β_1 = intercept and slope of the linear model

ε = error (*random*)

The simple linear regression model for the population

Simple linear regression model:

$$Y = \beta_0 + \beta_1 x + \varepsilon$$



The simple linear regression model for the population

Although it is assumed that a linear relationship exists between two variables at the population level, the **parameters β_0 and β_1 in the regression model** are **unknown**.

Sample data are therefore used to

- ▶ **Estimate** the parameters of the linear model
- ▶ Analytically **describe** the linear relationship between the dependent variable and the explanatory variable
- ▶ **Predict** the dependent variable based on the observed values of the explanatory variable using the estimated model

The simple linear regression model: sample data [READING]

- ▶ Simple linear regression model for the population:

$$Y = \beta_0 + \beta_1 x + \varepsilon \quad \text{with } \beta_0, \beta_1 \text{ not known}$$

- ▶ Simple linear regression model for the units in a random sample

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad i = 1, \dots, n$$

where:

x_i = value of the independent variable measured on the i -th sample observation

Y_i = random value of the dependent variable corresponding to x_i

ε_i = deviation of Y_i from the linear model (**random quantity**)

Linear model estimation: procedure [READING]

To estimate the linear model, we should:

- ▶ Propose **appropriate estimators** $\hat{\beta}_0$ and $\hat{\beta}_1$ for the unknown parameters, β_0 and β_1
- ▶ Use the sample data, $(x_1, y_1), \dots, (x_n, y_n)$ to obtain the **estimates**, b_0 and b_1 , i.e. the realisations of $\hat{\beta}_0$ and $\hat{\beta}_1$ based on the observed sample.

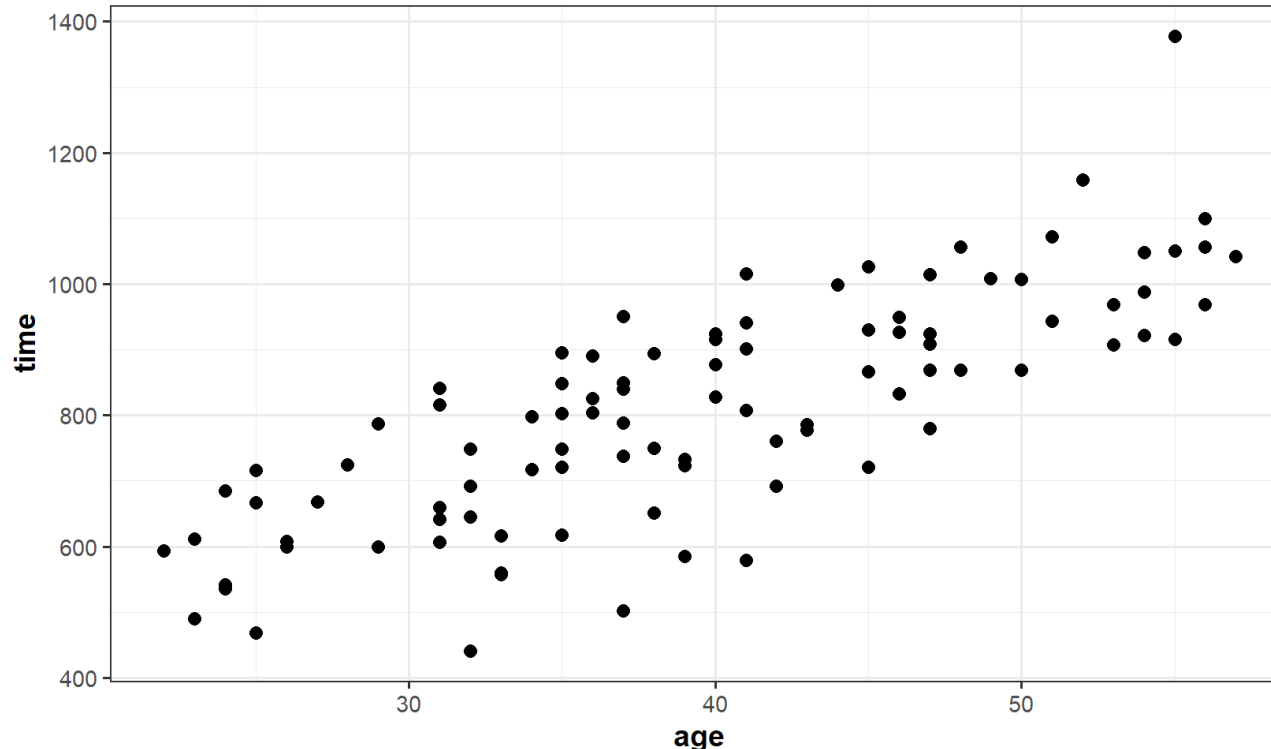
The procedure we follow to introduce the estimators used is slightly different:

- ▶ We consider the **sample data** and introduce the **least squares** approach to derive the parameters b_0 and b_1 of the **regression line** that **best interpolates the sample data** and introduce criteria to assess the quality of this line for the observed data
- ▶ We define the **least squares estimators** by generalising the estimates obtained on an observed sample to a random sample
- ▶ We study the **properties** of the **least-squares estimators** under appropriate assumptions

The regression line

Study of the relationship between time on chat and age (in years): on a sample of 96 users, information on certain socio-demographic characteristics is collected by telephone interview and the time (in minutes) spent on the site in a certain period is recorded from the site log data

Scatterplot:



At the sample level (descriptive), there is a relationship between the two variables, certainly a direct one.

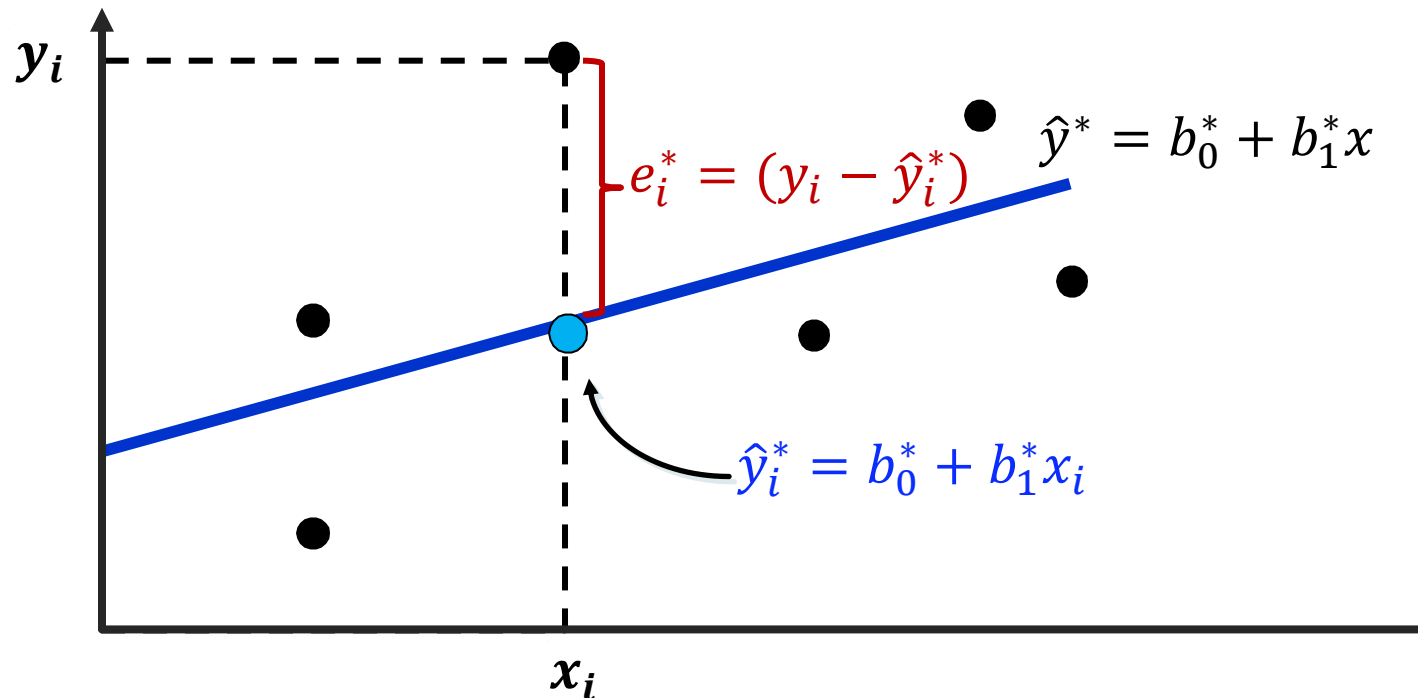
The linear approximation seems acceptable

The regression line: the least squares approach

Which line describes at best the linear relationship between two variables?

The **least squares approach** identifies the line that **cuts the data at best**

Consider a sample of data and a **generic straight line** with intercept b_0^* and slope b_1^*



For a sample observation characterised by (x_i, y_i) consider

- ▶ the prediction $\hat{y}_i^*$
- ▶ the deviation between the observed value and the value on the line, $e_i^* = y_i - \hat{y}_i^*$

Among all the possible straight lines, we choose the one **minimising the sum of the squared errors**

The regression line: the least squares approach

For each sample observation

x_i	observed value of X
y_i	value of Y observed at $X = x_i$
$\hat{y}_i^* = b_0^* + b_1^* x_i$	prediction of Y corresponding to $X = x_i$ using a straight line with intercept b_0^* and slope b_1^*

Consider the sum of the squared differences between the observed values $y_1, y_2, \dots, y_n$ and the predictions $\hat{y}_1^*, \hat{y}_2^*, \dots, \hat{y}_n^*$, i.e. the Sum of Squared Errors:

$$\text{SSE}^* = \sum_{i=1}^n (y_i - \hat{y}_i^*)^2 = \sum_{i=1}^n (y_i - b_0^* - b_1^* x_i)^2$$

We find the line that cuts the data at best by determining the coefficients minimising the sum of the squared differences between the observed and the predicted values

The regression line: the least squares approach

The coefficients that minimise the sum of the squared differences between the values observed on the dependent variable and the predictions obtained using a straight line can be determined by setting the partial derivatives equal to zero and solving the equations, obtaining:

$$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = \frac{s_{xy}}{s_x^2} = r_{xy} \frac{s_y}{s_x}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

And the equation of the least squares regression line is:

$$\hat{y}_i = b_0 + b_1 x_i$$

The regression line

Study of the relationship between time on chat and age. Assume that the sample statistics calculated from the collected data are: $\bar{x} = 39.4792$, $\bar{y} = 809.1354$, $s_x^2 = 88.06$, $s_y^2 = 30237.95$, $r_{xy} = 0.79$. What are the coefficients of the regression line? We can determine them 'by hand', using a calculator or using R as a calculator*:

$$s_{xy} = r_{xy} \sqrt{s_x^2 s_y^2} = 0.79 \sqrt{88.06 \cdot 30237.95} = 1289.118$$

$$b_1 = s_{xy} / s_x^2 = 1289.118 / 88.06 = 14.6391$$

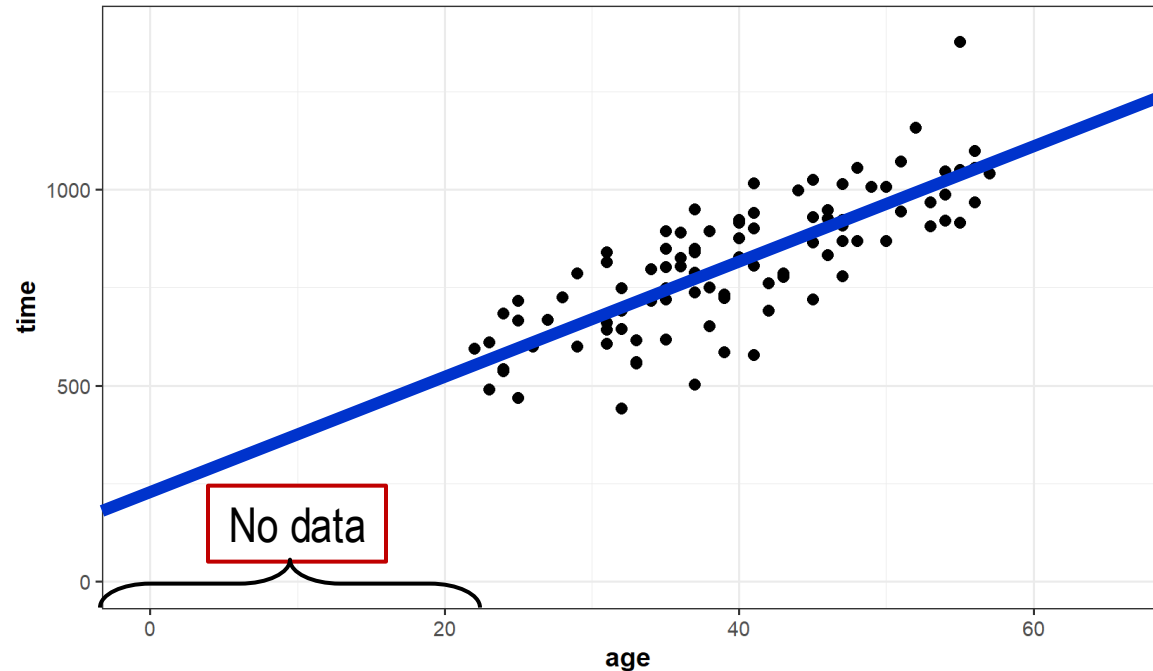
$$b_0 = \bar{y} - b_1 \cdot \bar{x} = 809.1354 - 14.6391 \cdot 39.4792 = 231.1954$$

 The equation of the line (rounding intermediate and final results) is:

$$\hat{y}_i = 231.20 + 14.64 \cdot x_i$$

* For the commands to obtain results using R see the script

Interpretation of the regression line



Estimated intercept, b_0

$$\hat{y}_i = 231.20 + 14.64 \cdot x_i$$

Estimated slope, b_1 .

Please note: b_0 cannot be interpreted as 'time spent on chat by subjects aged 0'. There is no information on subjects aged less than 20 (no predictions should be obtained corresponding to values outside the range of X).

For each additional year of age, based on the regression line, time spent on chat is 'expected' to increase on **average** by $b_1 = 14.64$ minutes (approximately, rounded coefficient)

Evaluation of the simple regression model [READING]

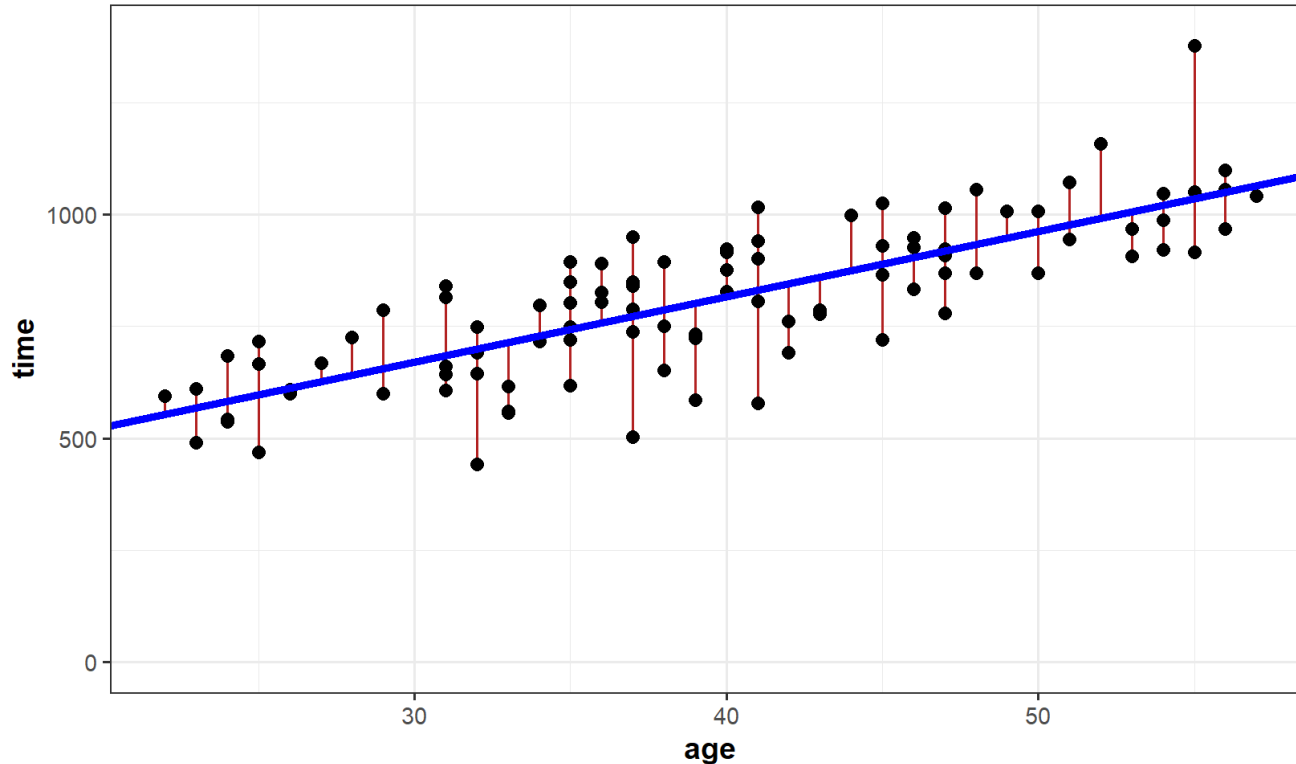
The least squares approach leads to the identification of the best line (with reference to the sum of the squared errors), *regardless of whether there is a relationship between the two variables or not*.

- ▶ It is therefore important to assess the adequacy of the model, i.e. the ability of the data collected on the explanatory variable to explain/recover the values observed on the dependent variable by means of a linear model based on the considered explanatory variable.
- ▶ To assess the goodness of fit of the regression line, we can refer at the sum of the squared **errors**, **SSE**, i.e. the sum of the squared differences between the observed values and those predicted using the regression line:

$$\text{SSE} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

Note: the mean of the residuals from the estimated least-squares model is 0 (the regression line cuts the data at best by balancing positive and negative deviations from the line)

The sum of squared errors, SSE



In this case the sum of the squared deviations of all the observed data (on the dependent variable, time) from the line is*:

$$SSE = 1079812$$

Since no assumptions were made about the amount of dispersion in data, it is difficult to evaluate **SSE** (as it was difficult to evaluate the value taken by the variance!), because it is an absolute measure whose value *also* depends on the unit of measurement of the dependent and explanatory variables. To assess the fit of the model, we therefore introduce a **relative** measure

* We will see later how the SSE can be calculated based on the available data

Goodness of fit of the regression line

The **SSE** is an **absolute measure**, whose value also depends on the units of measurement of the dependent variable and the explanatory variable.

To derive a **relative** measure (with a clear and well-defined range of variation), we introduce:

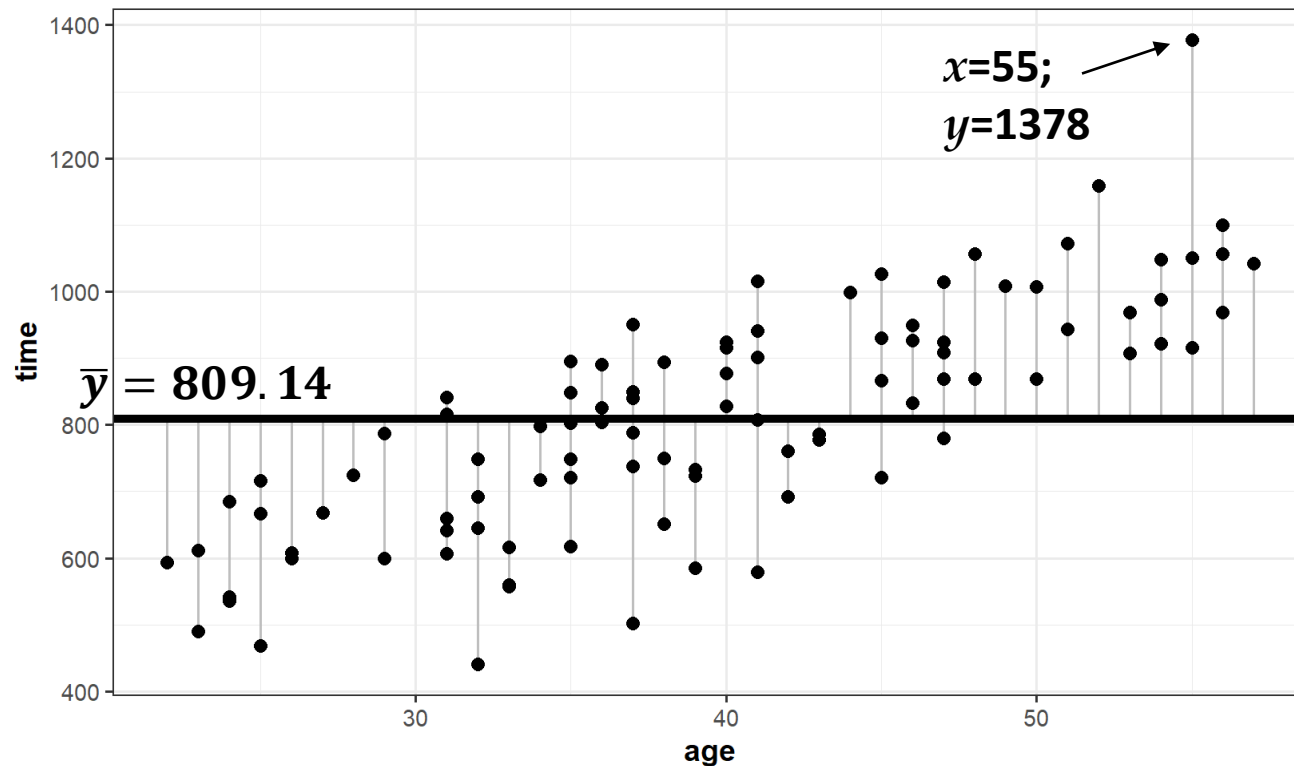
- ▶ The **Total Sum of Squares (SST)**, the sum of the squared deviations of the observed values $y_1, y_2, \dots, y_n$ from their mean, $\bar{y}$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = (n - 1)s_y^2$$

- ▶ The sum of the **squares of the regression (SSR)**, the sum of the squared deviations of the predictions $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$ from the mean of the dependent variable

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

The sum of the total sum of squares, SST

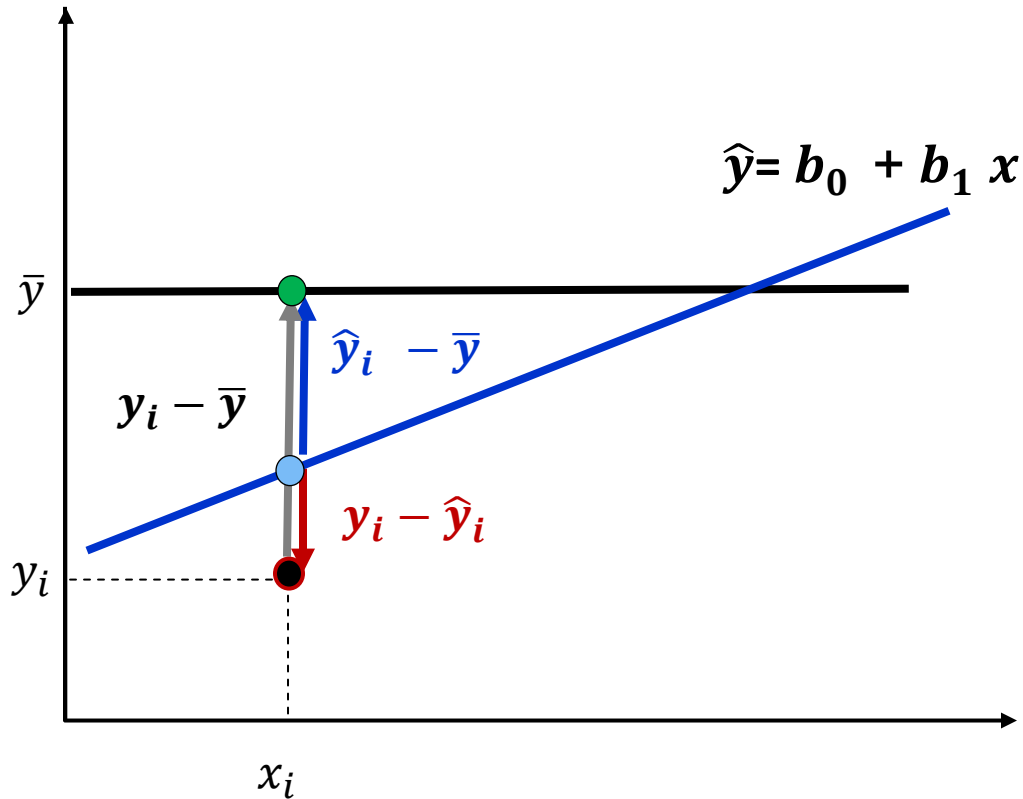


What if we were to predict the time spent on chat based only on the values taken by the dependent variable?

We would use the average of the observed values (809.1354)

The **SST** therefore gives an indication of the quality of the **mean** as a predictor of the dependent variable i.e. the fit of a straight line without a slope!!!

Decomposition of the total sum of squares



By referring to a single case, there is a relationship between the deviations.

What happens when we consider the sums of these deviations squared?

Decomposition of the total sum of squares

The following relationship can be proven between the sums of squares for n observations:

Total sum of squares

(Dispersion of y_i)

Regression sum of squares

(Dispersion explained by the line)

Error sum of squares

(Dispersion not explained by the line)

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

=

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

+

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

 $SST = SSR + SSE$

The coefficient of determination

By comparing **SSE** (or **SSR**) with its maximum value (**SST**), one can define a relative measure of the **goodness of fit** of the regression line, the so-called **coefficient of determination**:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

R^2 measures the proportion of the variation in the dependent variable explained by the regression line, and since $SST = SSE + SSR$, it takes on values between 0 and 1.

- ▶ $R^2 = 1$: The line perfectly explains the dependent variable (perfect linear relationship):
SSE = 0 : there are no prediction errors and the data are perfectly linearly aligned
- ▶ $R^2 = 0$: The line does not improve at all the prediction of the dependent variable compared to the mean.
SSR = 0: the regression line coincides with a line with zero slope and intercept equal to the mean of the dependent variable

R^2 seems strongly related to the **strength of the linear relationship** between the two variables. This is no coincidence: in fact, it can be shown that **in the case of simple linear regression** R^2 coincides with the **squared linear correlation coefficient** r_{xy}^2

The coefficient of determination

Note that:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = r_{xy}^2$$

Since:

$$SST = (n - 1)s_y^2$$

it is

$$SSR = SST \cdot r_{xy}^2 \text{ and } SSE = SST \cdot (1 - r_{xy}^2)$$

It is worth noting that (since $b_1 = s_{xy} / s_x^2$) :

$$SSR = SST \cdot r_{xy}^2 = (n - 1)s_y^2 \cdot r_{xy}^2 = (n - 1)b_1^2 s_x^2$$

Given the line, **SSR** increases as the variance of the explanatory variable increases!

The coefficient of determination

Study of the relationship between time on chat and age

Assess the goodness of fit of the estimated regression model:

$$\hat{y}_i = 231.20 + 14.64 \cdot x_i$$

$$r_{xy} = 0.79 \rightarrow R^2 = r_{xy}^2 = 0.79^2 = 0.6241$$

The 62% of the variation in time spent on chat is explained by age (thus, the time spent on chat changes depending on the user's age)

The remaining 38% is not explained by the model.

Age explains the 62% of the variability of the time on chat.

Simple linear model: from sample to population [READING]

Study of the relationship between time on chat and age

At the sample level, we determined the regression line

$$\hat{y}_i = 231.20 + 14.64 \cdot x_i, \text{ characterised by } R^2 = 0.6241$$

The line and its explanatory ability were determined on the basis of a specific **sample**. How can these results be extended to the **population**?

To evaluate the model with reference to the population, we must consider that the coefficients of the line are obtained by referring to a specific sample. In order to make inference about the linear model in the population, it is necessary to explicitly take into account the **sample variability**, thus defining **statistics** that can be used as **estimators or as test statistics** and allow for inference

Regression model: least-squares estimators [READING]

In order to estimate the coefficients of the linear model in the **population**, we propose the **least-squares estimators** obtained by substituting the specific sample observations, $y_1, y_2, \dots, y_n$, **with the random sample observations** $Y_1, Y_2, \dots, Y_n$, which describe the - random - measurements observed on a **generic random sample of values of the dependent variable** corresponding to the values observed on the explanatory variable, $x_1, x_2, \dots, x_n$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

In order to study the properties of these estimators, it is necessary - as always - to make **assumptions** on the characteristics of the distribution of Y or, equivalently, on the characteristics of the distribution of the unique random component of the linear model:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

that is the random error, ε

Simple regression

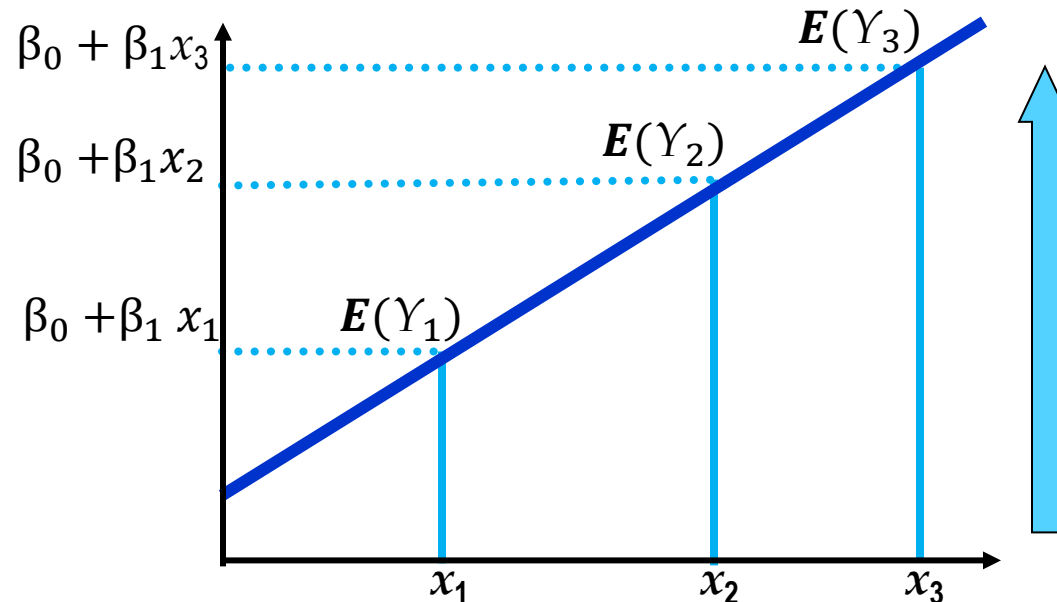
Weak and strong assumptions and inference on the linear model parameters

Weak assumptions

The error term is a r. v. with **expected value equal to zero**: $E(\varepsilon_i) = 0$ for each i

$$Y = \beta_0 + \beta_1 x + \varepsilon \quad \longrightarrow \quad E(Y_i) = \beta_0 + \beta_1 x_i \text{ for each } i$$

Thus, the information on X makes it possible to explain the **expected values** of Y_i (but **not** the specific realisations Y_i) and there are no other factors that systematically influence $E(Y_i)$



The expected value of Y varies linearly with X . It is assumed that there are no other factors that systematically influence Y i.e. that the differences in the expected values of Y are only due to X (or that X is sufficient to explain them)

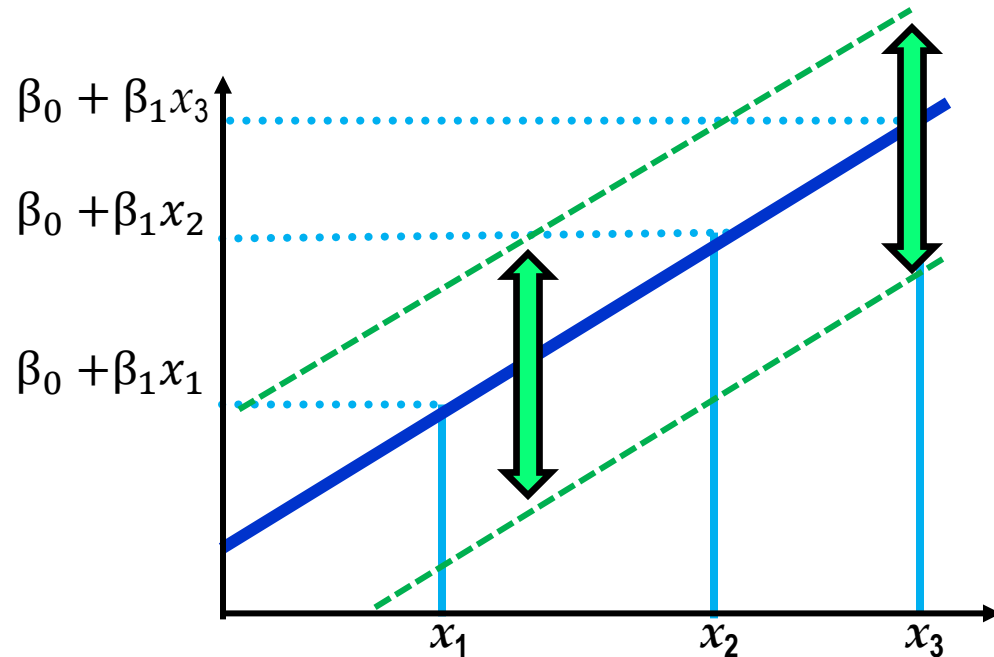
Weak assumptions

The error term is a r.v. with constant variance, σ_ε^2 whatever x_i (**homoskedasticity**):

$$\text{Var}(\varepsilon_i) = \sigma_\varepsilon^2 \text{ for every } i$$

$$Y = \beta_0 + \beta_1 x + \varepsilon \quad \longrightarrow \quad \text{Var}(Y_i) = \sigma_\varepsilon^2 \quad \text{for each } i$$

Thus, the variance (and standard deviation) of Y_i does not depend on x_i :



The standard deviation (the amount of dispersion around the line) remains constant as X changes. Thus, although they have different expected values, the Y_i have the same variance (they can be considered a homogeneous sample with reference to the squared deviation from the mean)

Weak assumptions

The errors $\varepsilon_1, \dots, \varepsilon_n$ are **uncorrelated**

$$Y = \beta_0 + \beta_1 x + \varepsilon \quad \longrightarrow \quad \begin{aligned} \text{Cor}(Y_i, Y_h) &= \text{Cor}(\beta_0 + \beta_1 x_i + \varepsilon_i, \beta_0 + \beta_1 x_h + \varepsilon_h) = \\ &= \text{Cor}(\varepsilon_i, \varepsilon_h) = 0 \text{ for each } i, h \end{aligned}$$

Thus, random variables - i.e. the sample realisations - $Y_1, \dots, Y_n$ are uncorrelated

Properties of least-squares estimators [OPTIONAL]

It can easily be proved that:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})}{(n-1)s_x^2} Y_i = \sum_{i=1}^n w_i Y_i$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x} = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} w_i \right) Y_i = \sum_{i=1}^n k_i Y_i$$

The least squares estimators are **linear combinations** of $Y_1, \dots, Y_n$

➡ It can be proved that under the weak assumptions the least squares estimators are uncorrelated, are unbiased for the corresponding parameters, and it is possible to derive their variance and their standard error

Moreover, when the sample size is large enough, the distributions of the least-squares estimators, which are in fact linear combinations of the Y_i , can be approximated by a normal.

Properties of least-squares estimators [IMPORTANT!]

It can be proved that **under the weak assumptions** the least squares estimators:

- ▶ Are **unbiased** for the population parameters

$$E(\hat{\beta}_1) = \beta_1$$

$$E(\hat{\beta}_0) = \beta_0$$

- ▶ Their variances **measure the expected squared deviation around the parameters of interest** :

$$Var(\hat{\beta}_0) = \sigma_\varepsilon^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2} \right)$$

$$Var(\hat{\beta}_1) = \frac{\sigma_\varepsilon^2}{(n-1)s_x^2}$$

As the sample size increases the variances (and therefore the standard errors) **of the estimators tend to zero**, and therefore the estimates of the coefficients are more and more concentrated around the corresponding population's parameters.

Standard error of the model

The **estimate** of the parameter σ_ε^2 at a specific sample realisation $y_1, \dots, y_n$ is:

$$s_\varepsilon^2 = \sum_{i=1}^n \frac{[y_i - (b_0 + b_1 x_i)]^2}{(n - 2)} = \frac{SSE}{(n - 2)} = MSE$$

also called the **mean of the squares of the errors (MSE)**, given by the **sum of the squares of the errors (SSE)** divided by $(n - 2)$

Thus, the deviations of the y_i observed around the estimated line are considered estimates of the model errors, and the **SSE** is used to estimate the dispersion.

The estimate of σ_ε is

$$s_\varepsilon = \sqrt{\frac{SSE}{(n - 2)}} \quad \text{standard error of the model or standard error of residuals}$$

Evaluation of point estimators

Study of the relationship between chat time and age.

Consider the estimated model:

$$\hat{y}_i = 231.20 + 14.64 \cdot x_i$$

Is it possible to assess the reliability of the estimated slope, 14.64? How would you proceed?

It is not possible to assess the reliability of the estimate but only of the **estimator**. The observed estimate is just one of the possible realisations of the estimator.

Remember that $n = 96$, $s_x^2 = 88.06$, $s_y^2 = 30237.95$, and $R^2 = 0.6241$. Therefore:

$$SSE = (1 - R^2) \cdot SST = (1 - R^2) \cdot (n - 1)s_y^2 = (1 - 0.6241) \cdot 95 \cdot 30237.95 = 1079812.31$$

$$s_\varepsilon^2 = SSE / (n - 2) = 1079812.314 / 94 = 11487.37$$

and:

$$se_{\hat{\beta}_1} = \sqrt{\frac{s_\varepsilon^2}{(n - 1)s_x^2}} = \sqrt{\frac{11487.37}{95 \cdot 88.06}} = 1.1718$$

The (estimate of the) expected deviation of a generic estimate of the slope from the parameter β_1 is **1.1718**

Inference on the model's parameters in the population

To apply more advanced inferential techniques, e.g. constructing confidence intervals or testing hypotheses, it is necessary to **determine the distribution of $\hat{\beta}_0$ and $\hat{\beta}_1$** .

For this purpose - at least when the sample size is not very large - it is necessary to make a further assumption on the **distribution of model errors**, in addition to the weak ones already assumed.

Weak assumptions: : $E(\varepsilon_i) = 0$ and $Var(\varepsilon_i) = \sigma_\varepsilon^2$ for each i ; $Cor(\varepsilon_i, \varepsilon_h) = 0$ for each i, h

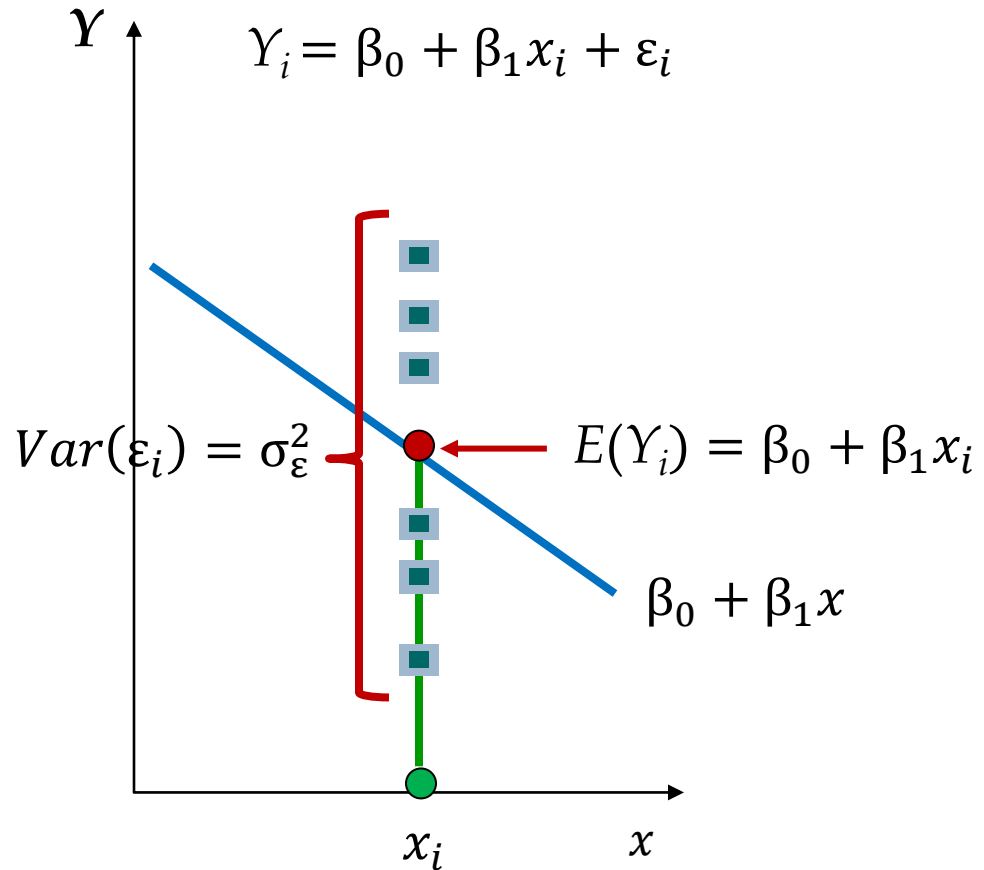
Strong assumptions: Weak assumptions + The error ε has normal distribution: $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$

$$Y = \beta_0 + \beta_1 x + \varepsilon \quad \longrightarrow \quad Y_i \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma_\varepsilon^2)$$

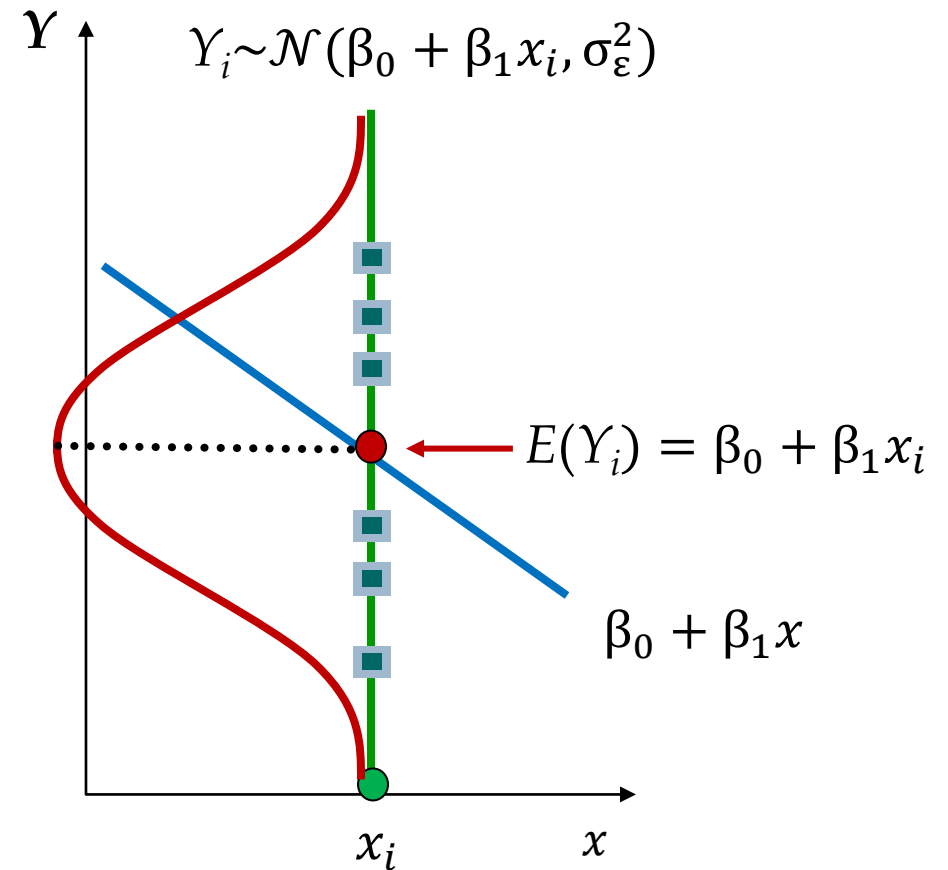
Under the normality hypothesis, non-correlation implies independence. Thus *under the strong assumptions* the Y_i are independent and normally distributed.

Strong Assumptions

Weak assumptions

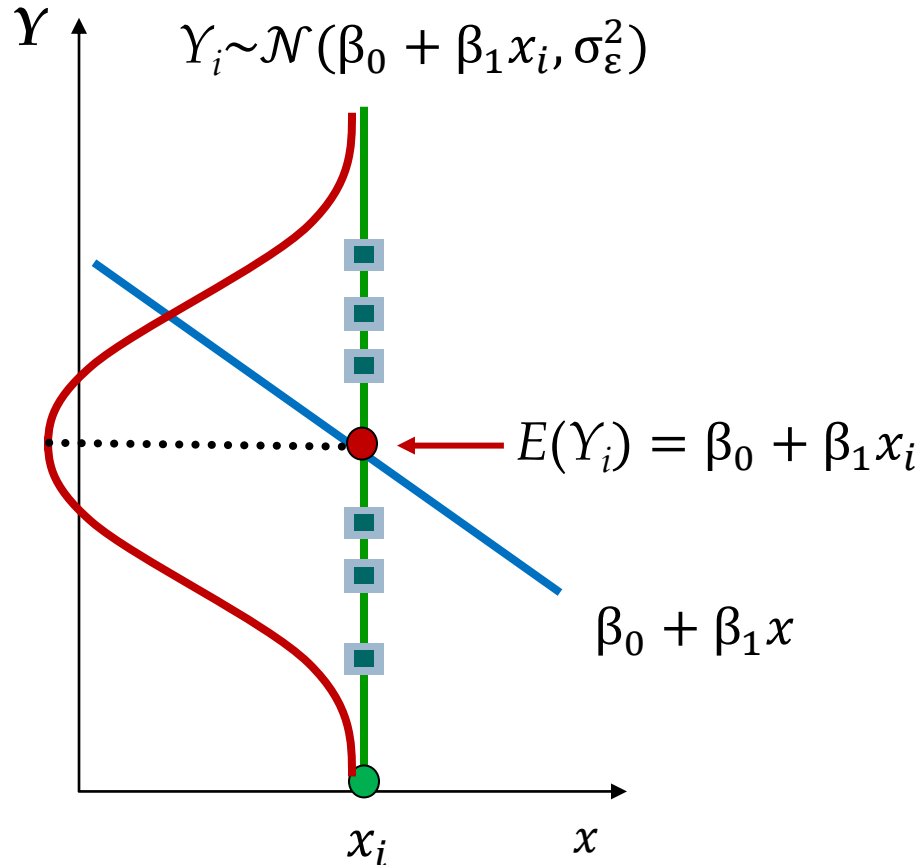


Strong assumptions



Strong Assumptions [READING]

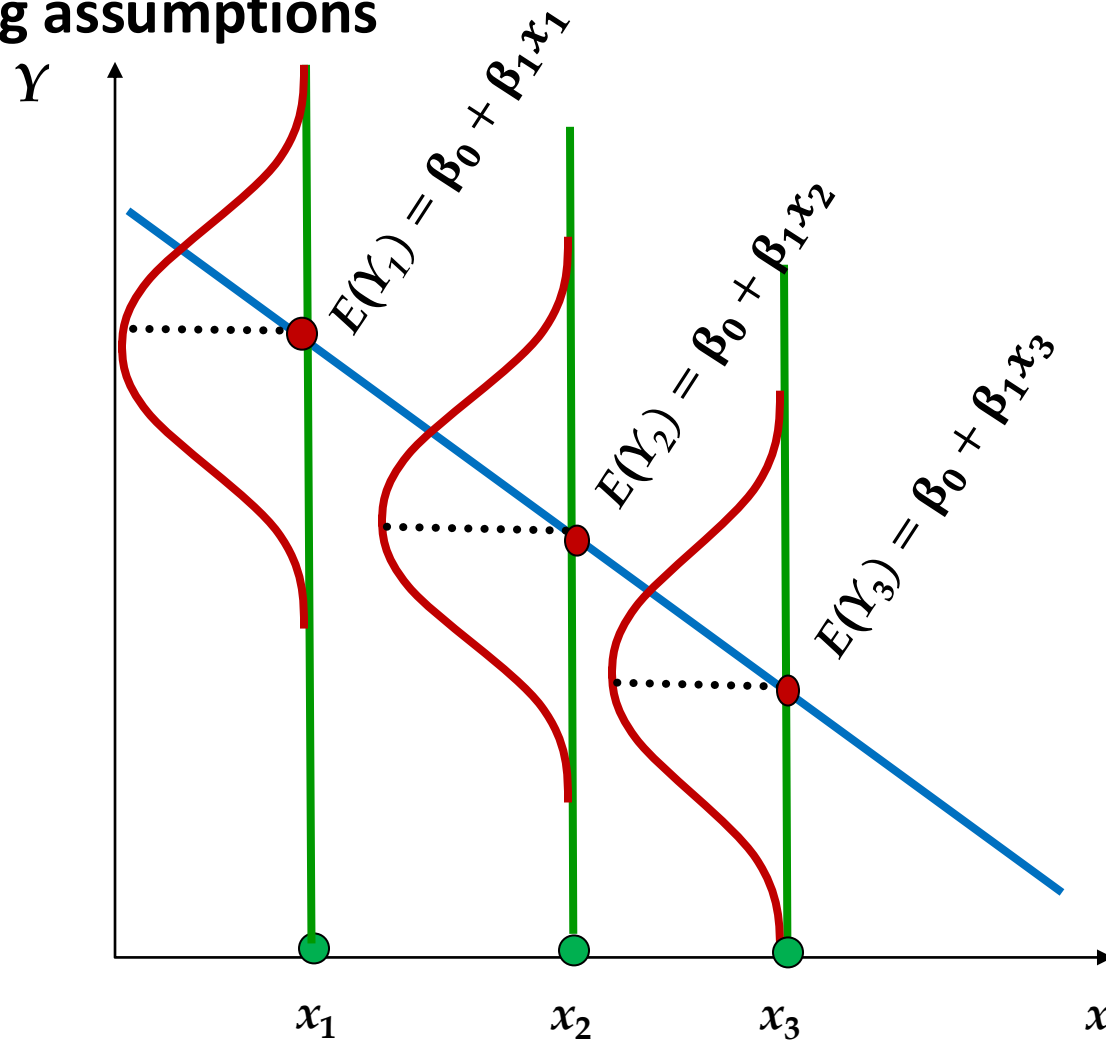
Strong assumptions



Y_i is dispersed around its expected value, $\beta_0 + \beta_1 x_i$ (the deterministic component of the linear model) with an expected absolute deviation equal to σ_ε whatever the value of x_i . As the values of the explanatory variable change, the distribution of Y_i **always has the same shape and only its expected value changes!!!!**

Strong Assumptions [READING]

Strong assumptions



Strong Assumptions [OPTIONAL]

In order to understand what are the implications of strong assumptions, consider that:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})}{(n-1)s_x^2} Y_i = \sum_{i=1}^n w_i Y_i$$

Known coefficient

The estimator of the slope is a linear combination of the Y_i , **which under the strong assumptions are independent and normally distributed**; therefore, **the estimator also has a normal distribution**, even when the sample size is limited

The same applies to the linear model's intercept estimator, $\hat{\beta}_0$.

Furthermore, under the strong assumptions the least squares estimators are independent.

Properties of least-squares estimators [RECAP]

The least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are both linear combinations of the Y_i .

Under **weak assumptions**:

$$E(\hat{\beta}_0) = \beta_0$$

$$Var(\hat{\beta}_0) = \sigma_\varepsilon^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2} \right)$$

$$E(\hat{\beta}_1) = \beta_1$$

$$Var(\hat{\beta}_1) = \frac{\sigma_\varepsilon^2}{(n-1)s_x^2}$$

Under **strong assumptions**:

$$\hat{\beta}_0 \sim \mathcal{N} \left(\beta_0, \sigma_\varepsilon^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2} \right) \right)$$

$$\hat{\beta}_1 \sim \mathcal{N} \left(\beta_1, \frac{\sigma_\varepsilon^2}{(n-1)s_x^2} \right)$$

Inference on β_1

We specifically refer to inference on β_1 , the parameter we are most interested in, since it measures the expected variation in the dependent variable corresponding to a variation in the explanatory variable (the same procedures can be used for inference on β_0).

Note that:

$$\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma_\varepsilon^2}{(n-1)s_x^2}\right) \longrightarrow \frac{\hat{\beta}_1 - \beta_1}{\frac{\sigma_\varepsilon}{\sqrt{(n-1)s_x^2}}} = \frac{\hat{\beta}_1 - \beta_1}{SE_{\hat{\beta}_1}} \sim \mathcal{N}(0,1)$$

This result cannot be used because σ_ε^2 is not known!

Replacing σ_ε , unknown, with its estimator, we obtain a statistic that (under the strong assumptions) has a Student's t distribution with $(n-2)$ degrees of freedom

$$\frac{\hat{\beta}_1 - \beta_1}{\frac{\hat{\sigma}_\varepsilon}{\sqrt{(n-1)s_x^2}}} = \frac{\hat{\beta}_1 - \beta_1}{\widehat{SE}_{\hat{\beta}_1}} \sim t_{n-2}$$

Confidence interval for β_1

Since

$$\frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}_\varepsilon \sqrt{\frac{1}{(n-1)s_x^2}}} = \frac{\hat{\beta}_1 - \beta_1}{\widehat{SE}_{\hat{\beta}_1}} \sim t_{n-2}$$

The confidence interval estimate at the level $(1 - \alpha)$ for β_1 based on the estimate of the slope determined on the basis of a specific sample (i.e. the sample realisation of $\hat{\beta}_1$), b_1 , is:

$$ci_{1-\alpha}(\beta_1) = [b_1 \mp t_{n-2;\alpha/2} \cdot se_{\hat{\beta}_1}] = \left[b_1 \mp t_{n-2;\alpha/2} \cdot s_\varepsilon \sqrt{\frac{1}{(n-1)s_x^2}} \right]$$

where $t_{n-2,\alpha/2}$ is the percentile of order $(1 - \alpha/2)$ of a **Student's t** distribution with $(n - 2)$ degrees of freedom.

Testing Hypotheses on β_1

It is possible to use the illustrated results to test the null hypothesis

$$H_0 : \beta_1 = \beta_1^0$$

Against alternative hypotheses such as:

$$H_1: \beta_1 < \beta_1^0 \text{ or } H_1: \beta_1 \neq \beta_1^0 \text{ or } H_1: \beta_1 > \beta_1^0$$

The test statistic to be used is

$$\frac{\hat{\beta}_1 - \beta_1^0}{\hat{\sigma}_\varepsilon \sqrt{\frac{1}{(n-1)s_x^2}}} = \frac{\hat{\beta}_1 - \beta_1^0}{\widehat{SE}_{\hat{\beta}_1}}$$

which under the null hypothesis has a **Student's t** distribution with $(n - 2)$ **degrees of freedom**

Testing Hypotheses on β_1

Given a level of significance, α , the rejection region of the null hypothesis is built by comparing the value of the test statistic observed in a sample:

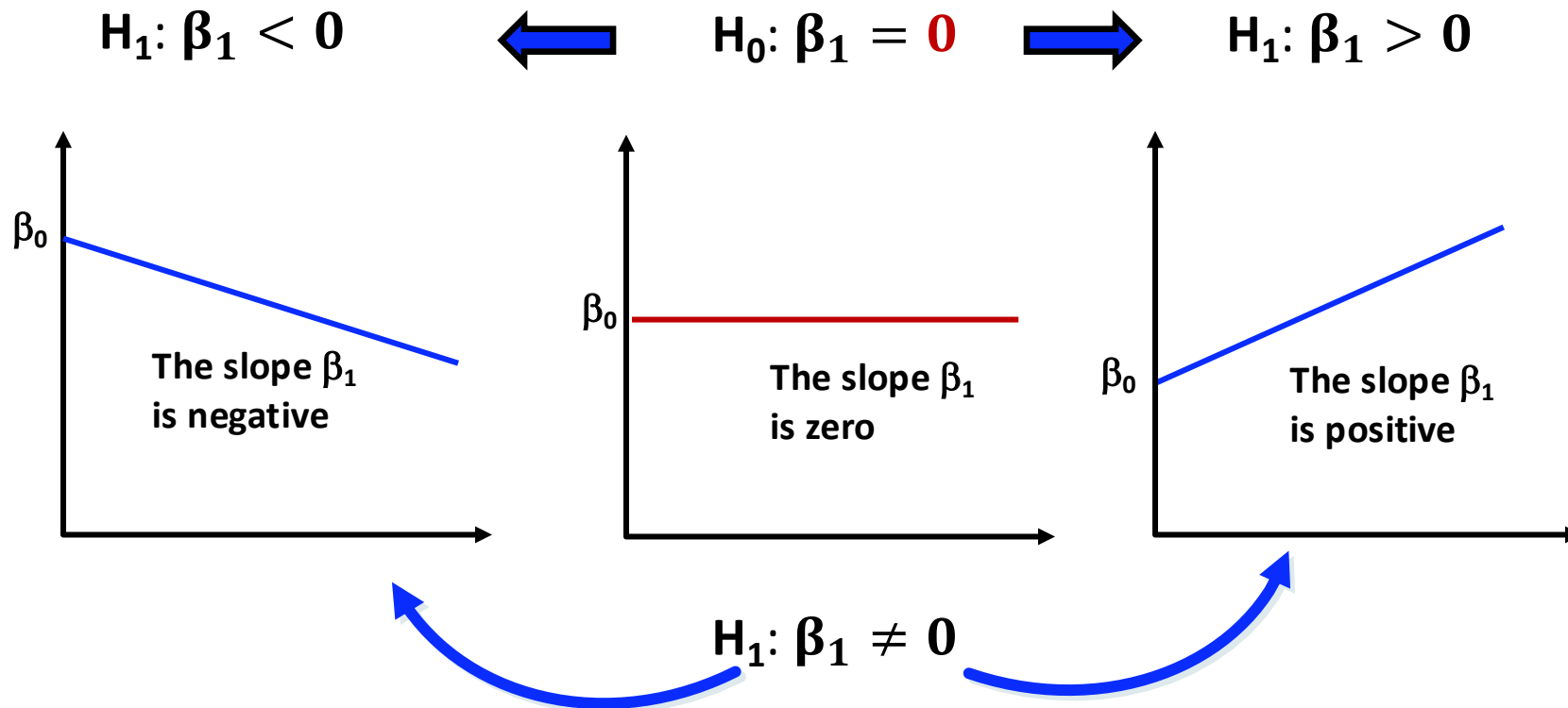
$$t_{obs} = \frac{b_1 - \beta_1^0}{s_\varepsilon \sqrt{\frac{1}{(n-1)s_x^2}}} = \frac{b_1 - \beta_1^0}{se_{\hat{\beta}_1}}$$

with the appropriate percentile of **Student's t** distribution with $(n - 2)$ degrees of freedom :

$$H_0: \beta_1 = \beta_1^0 \quad vs \quad \begin{cases} H_1: \beta_1 < \beta_1^0 & \Rightarrow R = \{t_{obs} < -t_{(n-2),\alpha}\} \\ H_1: \beta_1 \neq \beta_1^0 & \Rightarrow R = \{|t_{obs}| > t_{(n-2),\alpha/2}\} \\ H_1: \beta_1 > \beta_1^0 & \Rightarrow R = \{t_{obs} > t_{(n-2),\alpha}\} \end{cases}$$

Testing Hypotheses on β_1

Usually, we are interested in testing the null hypothesis that the parameter β_1 is equal to zero, i.e. that the line has zero slope in the population and that the linear model defined on the basis of the considered explanatory variable is ineffective, towards the alternative that the coefficient is non-zero.



Testing Hypotheses on β_1

Study of the relationship between time on chat and age.

Check whether there is sufficient empirical evidence to conclude that the slope of the regression model is different from 0 (in the population), using a test of level $\alpha = 0.05$.

We must test the hypotheses: $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$

Rejection region?

The realisation of the test statistic is:

$$t_{obs} = \frac{b_1 - 0}{s_\varepsilon \sqrt{\frac{1}{(n-1)s_x^2}}} = \frac{b_1}{se_{\hat{\beta}_1}}$$

Since we are testing a simple hypothesis against a bilateral the rejection region is:

 **Region of rejection : $|t_{obs}| > t_{n-2; \alpha/2}$**

Testing Hypotheses on β_1

Study of the relationship between time on chat and age

Rejection region to test $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$ at level $\alpha = 0.05$.

Since $n = 96$, the critical value of the rejection region (percentile of the t distribution) is

$t_{94;0.025} = 1.9855$:

```
> qt(0.975, df=94)
[1] 1.985523
```

and the rejection region is

$$R = \left\{ t_{obs} = \left| \frac{b_1}{se_{\hat{\beta}_1}} \right| > 1.986 \right\}$$

Remember that $b_1 = 14.64$ and $se_{\hat{\beta}_1} = 1.1718$ (rounded results). Therefore:

$$t_{obs} = \frac{14.64}{1.1718} = 12.49$$

Since $t_{obs} = 12.49 > 1.986$, H_0 is rejected in favour of the alternative: there is sufficient evidence to conclude that the slope of the linear model in the population is different from zero

Testing Hypotheses on β_1

Study of the relationship between time on chat and age.

What is the p-value of the observed sample realisation?

The p-value of the sample realisation is the probability of observing a value of the test statistic less favourable to the null hypothesis H_0 than the observed one:

$$\text{P-value} = 2Pr\left(t_{n-2} > \left| \frac{b_1}{se_{\hat{\beta}_1}} \right| \right) = 2Pr(t_{n-2} > 12.49)$$

```
> 2*(1-pt(12.49,df=94))  
[1] 0
```

Interpretation? Conclusions?

The observed sample realisation is extremely 'far' from H_0 in probabilistic terms: the probability of observing a sample regression line with such a high slope when the population is actually characterised by a regression line with zero slope is practically zero. We underline that the rejection of the null hypothesis does not depend on the chosen significance level, being the p-value lower than any standard value of α

Confidence intervals for β_1

Study of the relationship between time on chat and age.

What is the confidence interval at the 0.95 level for β_1 ?

The estimated confidence interval is

$$\begin{aligned} ci_{1-\alpha}(\beta_1) &= [b_1 \mp t_{n-2;0.025} \cdot se_{\hat{\beta}_1}] = [14.64 \mp 1.9855 \cdot 1.1718] \\ &= [12.31, 16.97] \end{aligned}$$

Interpretation?

Since the 95% of the estimated intervals (based on all the possible samples of size 96 that can be drawn from the population) contain the parameter or, in other words, the probability that a generic interval contains the parameter is 0.95, one can **expect** with 95% confidence a variation in the **average** time spent on the chat between 12.31 and 16.97 minutes corresponding to 1-year variation in the age of a user

Cautions in drawing conclusions based on inference on β_1

Rejecting the hypothesis that $\beta_1 = 0$ and concluding that the slope of the line is not zero and that the estimated slope is statistically significant

- ▶ **does not allow** to infer that there is a cause-and-effect relationship between X and Y . This conclusion is only possible when there is some theoretical justification to support it.
- ▶ **does not allow to** conclude that the relationship between X and Y is **linear**. One can only say that **for values of X in the range of the observed sample values of X (i.e. values not too dissimilar/distant from those on which the estimate was based on)** a significant linear variation of the dependent variable can be expected corresponding to a 1-unit variation of the explanatory variable.
- ▶ **does not allow** to conclude that the linear model explains to a satisfactory extent the dependent variable. This conclusion can be drawn only based on the determination coefficient, but still even if such coefficient is high it is not possible to state that the relation is linear. One can only say that the linear model based on X explains a significant portion of the variance of Y **for values of X in the range of the sample values of X (i.e. values not too dissimilar/distant from those on which the estimate was based on)**

Simple regression

Prediction

Prediction

The estimated regression model can be used to make predictions about the dependent variable at a given value of the explanatory variable.

As we have already pointed out, before **using the model** it would be appropriate to check whether the assumptions (weak and strong) at the basis of the linear model are indeed reasonable. However, we postpone this argument even though it is fundamental.

Let us consider a new observation characterised by a value of X equal to x_g . According to the linear model:

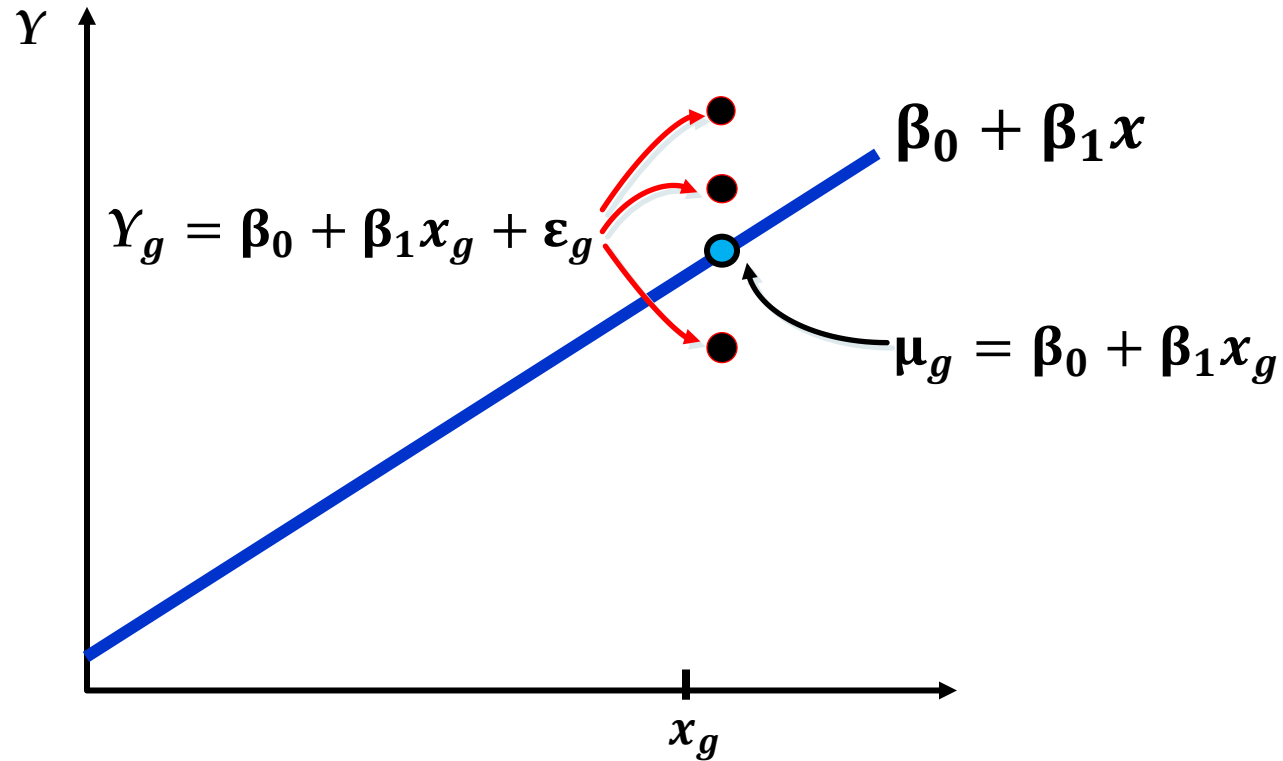
$$Y_g = \beta_0 + \beta_1 x_g + \varepsilon_g \quad \longrightarrow \quad E(Y_g) = \mu_g = \beta_0 + \beta_1 x_g$$

It is important to distinguish between:

- ▶ The prediction of the **expected value** of Y corresponding to x_g , $\mu_g = E(Y_g)$
- ▶ The prediction of the **exact value** of Y corresponding to x_g , Y_g . Compared to μ_g , Y_g also 'incorporates' the unobservable error ε_g , which causes the deviation of Y_g from the regression line (i.e. from μ_g).

Point Prediction

The two types of prediction



Point Prediction

- ▶ **The estimator of $E(Y_g) = \mu_g$.** To a specific value of x_g , corresponds a **sub-population** of values of Y whose average is estimated with:

$$\hat{\mu}_g = \hat{\beta}_0 + \hat{\beta}_1 x_g$$

- ▶ **The estimator of $Y_g = \mu_g + \varepsilon_g$.** Since the erratic component is random and cannot be estimated, the specific value of Y corresponding to x_g is estimated using the same **point estimator** used to predict μ_g :

$$\hat{Y}_g = \hat{\beta}_0 + \hat{\beta}_1 x_g$$

The (identical) point estimates are obtained by substituting the coefficient estimators with the estimates based on the available sample:

$$\hat{\mu}_g = \hat{y}_g = b_0 + b_1 x_g$$

Point Prediction

Study of the relationship between time on chat and age.

Estimated regression line (rounded values)

$$\hat{y}_i = 231.20 + 14.64 \cdot x_i$$

- ▶ **What is the average time spent in chat by a generic 30-year-old user?**

$$\hat{\mu}_g = 231.20 + 14.64 \cdot 30 = 670.4$$

- ▶ **What is the time spent on chat by a 30-year-old user?**

The point estimate is the same:

$$\hat{y}_g = 231.20 + 14.64 \cdot 30 = 670.4$$

However, the prediction of the time spent on chat by a *user aged 30* (i.e. not the average time for all users the users aged 30, but the time spent by a generic specific user) should include the error term, which, however, cannot be predicted

Confidence and prediction intervals

Rather than a point estimator, it is recommendable to build confidence intervals for predictions, that **account for the uncertainty on prediction**. Even if the point estimators for $E(Y_g) = \mu_g$ and Y_g are the same, the intervals built for the two parameters are different

- ▶ **Confidence interval** for μ_g , the expected value of Y corresponding to x_g :

$$ci_{1-\alpha}(\mu_g) = \left[\hat{y}_g \mp t_{(n-2),\alpha/2} s_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1)s_x^2}} \right]$$

- ▶ **Prediction interval** for Y_g , the value of Y predicted at x_g :

$$ci_{1-\alpha}(Y_g) = \left[\hat{y}_g \mp t_{(n-2),\alpha/2} s_\varepsilon \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1)s_x^2}} \right]$$

The prediction interval is wider than the confidence interval, since besides accounting for the uncertainty due to the estimation of the expected value, it also accounts for uncertainty due to the deviation of individual values from the expected value

The effect of x_g on the width of the intervals

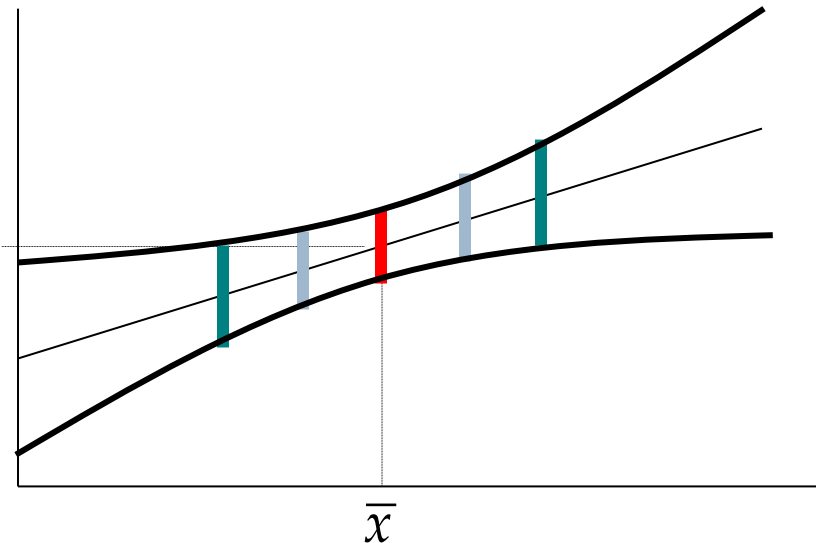
Confidence interval for $\mu_g = E(Y_g)$

$$\hat{y}_g \pm t_{(n-2), \alpha/2} S_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1)s_x^2}}$$

Forecast interval for Y_g

$$\hat{y}_g \pm t_{(n-2), \alpha/2} S_\varepsilon \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1)s_x^2}}$$

The width of the intervals increases with the distance of x_g from $\bar{x}$, the mean of the values of the explanatory variable calculated on the sample data used to estimate the linear model



Intervals have minimum width when $x_g = \bar{x}$; the width increases as x_g deviates from $\bar{x}$.

This is because when x_g is far from $\bar{x}$ predictions rely on values of $\mathbf{X}$ that are far from those used to estimate the model (risk associated with extrapolation)

Confidence and prediction intervals

Study of the relationship between time on chat and age.

95% confidence interval for the average time spent on chat by 30-year-old users?

$$ci_{1-\alpha}(\mu_g) = \left[\hat{y}_g \mp t_{(n-2), \alpha/2} s_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1)s_x^2}} \right]$$

Remember that $n = 96$, $s_x^2 = 88.06$, $\bar{x} = 39.4792$, $s_\varepsilon^2 = 11487.37$, $t_{94;0.025} = 1.9855$, and $\hat{y}_g = 670.4$.

Therefore:

$$\begin{aligned} ci_{1-\alpha}(\mu_g) &= \left[670.4 \mp 1.9855 \sqrt{11487.37 \cdot \left(\frac{1}{96} + \frac{(30 - 39.4782)^2}{(96 - 1) \cdot 88.06} \right)} \right] \\ &= [670.4 \mp 30.954] = [639.45, 701.35] \end{aligned}$$

Confidence and prediction intervals

Study of the relationship between time on chat and age.

95% confidence interval for the time spent in chat by a 30-year-old user?

$$ci_{1-\alpha}(Y_g) = \left[\hat{y}_g \mp t_{(n-2), \alpha/2} s_\varepsilon \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1)s_x^2}} \right]$$

Remember that $n = 96$, $s_x^2 = 88.06$, $\bar{x} = 39.4792$, $s_\varepsilon^2 = 11487.37$, $t_{94;0.025} = 1.9855$, e $\hat{y}_g = 670.4$.

Therefore:

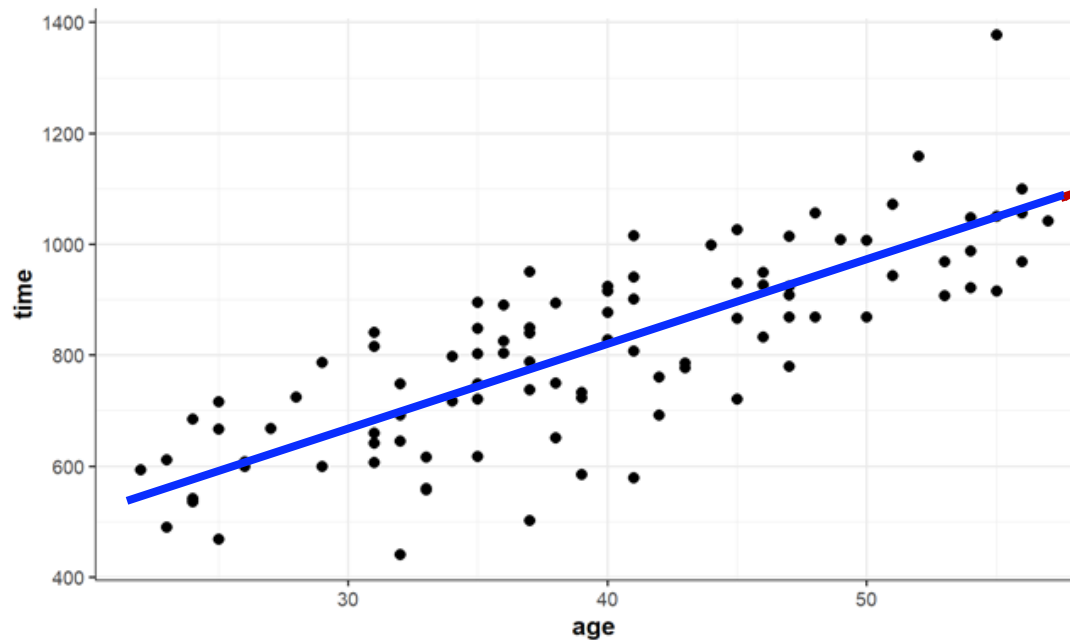
$$\begin{aligned} ci_{1-\alpha}(\mu_g) &= \left[670.4 \mp 1.9855 \sqrt{11487.37 \cdot \left(1 + \frac{1}{96} + \frac{(30 - 39.4782)^2}{(96 - 1) \cdot 88.06} \right)} \right] \\ &= [670.4 \mp 215.04] = [445.36, 885.44] \end{aligned}$$

The confidence interval for Y_g is **much** wider than that one determined for μ_g

Extrapolation [READING]

Study of the relationship between time on chat and age.

**Would you use the estimated model to predict the time spent on chat by a user aged 90?
Why?**

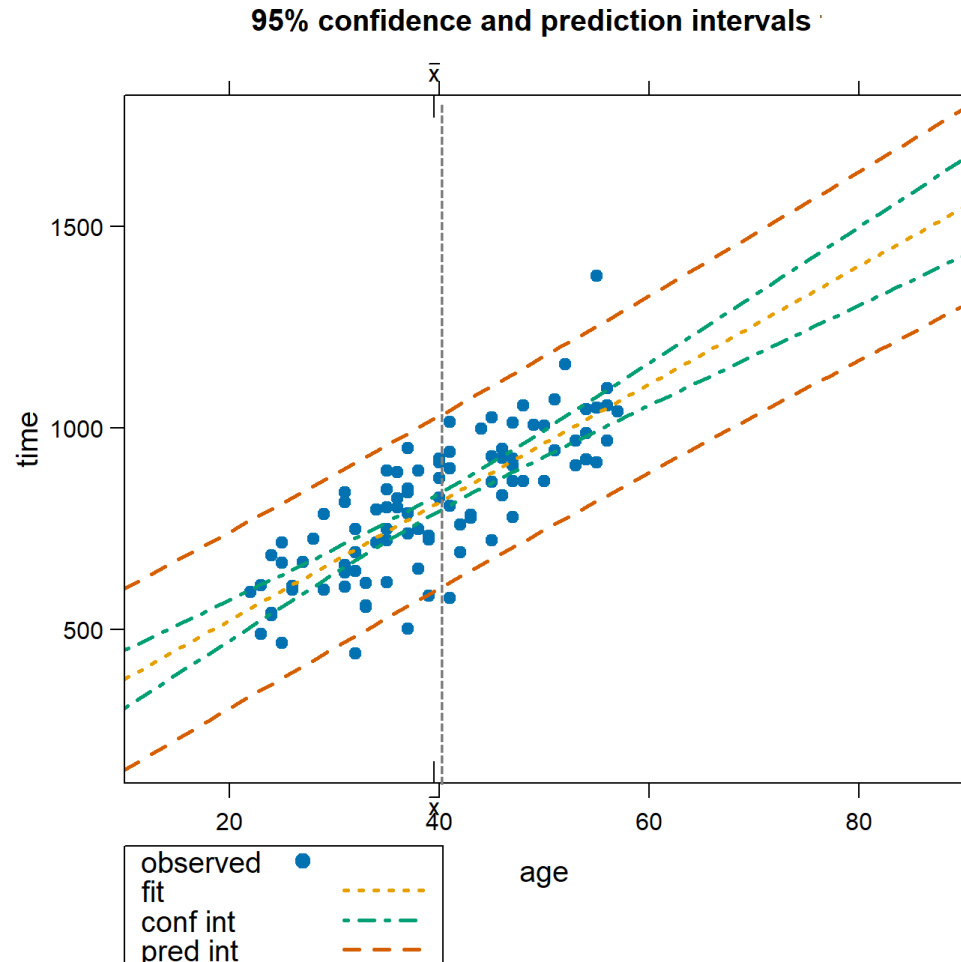


The model was estimated based on a sample of users with a maximum age of 60.

Extrapolating and obtaining predictions for users aged much more than 60 is very risky as we have no data to support the assumption that the trend observed for age values in the range of X remains the same even for age values far outside the range.

Extrapolation [READING]

Study of the relationship between time on chat and age.



Note how the width of the confidence and prediction interval increases as x_g moves away from the mean, $\bar{x}$. This reflects the risk associated with predictions far from the 'centre' of the data used to estimate the model. Note in particular that the impact of the distance of x_g from $\bar{x}$ is much less relevant for the prediction interval. Indeed, the third addendum in $1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1)s_x^2}$ is typically much lower than 1, and therefore it becomes relevant only when x_g is very far from $\bar{x}$

Rstudio: `lm()` function for estimating the linear model

The function `lm(formula, data)` in R allows estimating a linear model based on raw data.

In the function:

`formula` is a description of the model to be estimated, which in the case of a **simple linear regression** model has the form

$$\text{formula} = y \sim x$$

Where **`y`** is the dependent variable and **`x`** is the explanatory variable.

In the formula, **`x`** and **`y`** may be two vectors or the names of two variables contained in the dataframe specified in **`data`**.

Linear model estimation

The dataframe `Basket`* contains information on the customers of a supermarket who live in a specific area of a large city. In particular, besides other information, an indicator of household economic well-being (called `income` for simplicity) is obtained based on the declared address, the price per square metre of dwellings and rental prices, and the declared number of cohabitants. For each customer, the 'quality' of their expenditure is assessed on the basis of the products purchased (`basket`), taking into account the prices and alternatives available. One wants to assess whether the `income` indicator is related to the expenditure quality indicator, `basket`.

To estimate the linear model: `lm(formula = basket ~ income, data = Basket)`
`lm(formula = Basket$basket ~ Basket$income)`

```
> lm(formula = basket ~ income, data = Basket)
```

Call:

```
lm(formula = basket ~ income, data = Basket)
```

Coefficients:

```
(Intercept) income  
3.16383 0.01097
```

Note that by giving the command as an expression, the function only returns the coefficients estimates

Linear model estimation

It is convenient to assign the output obtained by applying the `lm()` function to an R object. The saved object is a **list**

```
> sreg.model <- lm(basket ~ income, data = Basket)
> str(sreg.model)
List of 12
 $ coefficients : Named num [1:2] 3.164 0.011
 ..- attr(*, "names")= chr [1:2] "(Intercept)" "income"
 $ residuals : Named num [1:99] 4.81 1.3 3.79 -1.42 -2.68 ...
 ..- attr(*, "names")= chr [1:99] "1" "2" "3" "4" ...
 $ effects : Named num [1:99] -669.31 -166.19 3.44 -1.87 -3.04 ...
 ..- attr(*, "names")= chr [1:99] "(Intercept)" "income" "" ...
 $ rank : int 2
 $ fitted.values: Named num [1:99] 37.8 70.1 46.1 65.3 48.2 ...
 ..- attr(*, "names")= chr [1:99] "1" "2" "3" "4" ...
 ..
 $ model : 'data.frame': 99 obs. of 2 variables:
 ..$ basket: num [1:99] 42.6 71.4 49.9 63.9 45.5 ...
 ..$ income: num [1:99] 3156 6101 3914 5662 4103 ...
 ..- attr(*, "terms")=Classes 'terms', 'formula' language basket ~ income
```

A list is an object whose elements may be objects of **any** type. The elements and sub-elements of a list can be accessed using the `$` symbol (e.g. `sreg.model$coefficients` as well as `sreg.model$model$basket`)

Linear model estimation

To display more detailed results, the `summary()` function can be applied to the output of the `lm()` function

```
> sreg.model <- lm(basket ~ income, data = Basket)
> summary(sreg.model)
```

Call:

`lm(formula = basket ~ income, data = Basket)`

Residuals:

Min	1Q	Median	3Q	Max
-22.0429	-7.2305	0.0133	6.3991	23.6122

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1638282	3.9037148	0.81	0.42
income	0.0109748	0.0006467	16.97	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.793 on 97 degrees of freedom

Multiple R-squared: 0.748, Adjusted R-squared: 0.7454

F-statistic: 288 on 1 and 97 DF, p-value: < 2.2e-16

$se_{\hat{\beta}_0}, se_{\hat{\beta}_1}$

$b_1 / se_{\hat{\beta}_1}$

??

s_ε

R^2

Confidence intervals for coefficients with R/RStudio

To determine the confidence intervals of the coefficients of the linear model, it is possible to apply the function

```
confint(object, level=0.95)
```

where `object` is the output of the `lm()` function, and `level` is the confidence level of the interval

99% confidence intervals for the coefficients of the linear model?

```
> sreg.model$coefficient
(Intercept) income
 3.16382818 0.01097483
>
> confint(sreg.model, level=0.99)
              0.5 %          99.5 %
(Intercept) -7.093056538 13.42071290
income       0.009275569  0.01267409
```

Confidence intervals for coefficients with R/RStudio

```
> sreg.model$coefficient
(Intercept)      income
  3.16382818  0.01097483
>
> confint(sreg.model, level=0.99)
              0.5 %      99.5 %
(Intercept) -7.093056538 13.42071290
income       0.009275569  0.01267409
```

*The coefficient of **income** is close to zero:
can we conclude that **income** is not
particularly relevant in the explanation of
basket?*

The statement is completely **wrong**!

- 1) The p-value of the slope ($\cong 0$) leads to **reject the null hypothesis that the coefficient is zero**.
- 2) R^2 indicates that a high proportion (about 75%) of the **basket** variation is explained by differences in customer **income**.
- 3) The estimated coefficient, 0.009276, represents the **average** (estimated) **basket** change corresponding to a unit change in **income**.

Indeed* for **basket** it is $\bar{y}=67.27$ and $s_y=19.41$, while for **income** $\bar{x}=5841.07$ e $s_x=1529.66$!!!

At a 'standard' variation of **income** we expect a variation of $0.009276 \cdot 1529.66 \cong 14.188$, which - compared to the standard deviation of **basket** - is not small at all!!! **The unit of measurement of b_1 depends on the units of measurement of X and Y !**

Prediction with R/RStudio

To determine the confidence intervals for the mean of the prediction and the forecast interval, you can use the function

```
predict(object, newdata, interval, level=0.95)
```

where

`object` is the output of the `lm()` function

`newdata` is a **dataframe** containing the values of the explanatory variable at which the prediction is to be determined. **The name of the variable in the dataframe must match the name of the variable in the dataframe used to estimate the model.**

`interval` is the type of interval required; you can specify '`none`' to get only the point forecast, '`confidence`' to get the confidence interval for the expected value, and '`prediction`' to get the prediction interval

`level` is the confidence level of the interval (default=0.95)

Forecasting with RStudio

Relationship between **income** and quality of expenditure (**basket**).

Obtain the 95% intervals for the prediction of the average **basket** value for two customers with **income** levels of 7500 and 9000 respectively

```
> # dataframe creation
> newdata<-data.frame(income=c(7500,9000))
> newdata
  income
1   7500
2   9000
> predict(sreg.model,newdata=newdata,
          interval='confidence',level=0.95)
      fit      lwr      upr
1  85.47505 82.58536 88.36475
2 101.93730 97.43652 106.43808
> predict(sreg.model,newdata=newdata,
          interval="prediction",level=0.95)
      fit      lwr      upr
1  85.47505 65.82438 105.1257
2 101.93730 81.98597 121.8886
```

Interpretation?

Simple regression

Analysis of residuals

Analysis of residuals: motivation

Even before using the linear model to make predictions, one should assess whether the assumptions necessary to make inferences about the model are actually fulfilled.

There are various procedures and even specific tests developed for this purpose. However, **we follow an exploratory approach, and describe how to use regression residuals to assess:**

- ▶ Whether the assumption of **linearity** is supported by the data (not particularly useful in the case of simple linear regression because the relation can be visualized using a scatterplot; we see it here to better understand its use in the case of multiple linear regression).
- ▶ If there are systematic structural differences between the **average levels** of error at specific values of the explanatory variable
- ▶ If the variance of the errors is indeed constant (assumption of **homoskedasticity**)
- ▶ If errors have normal **distribution***
- ▶ If there are specific cases that overly influence the results (**outliers**)
- ▶ The possible **correlation** between errors; this can be a problem in specific applications (surveys over time, surveys on the same units under different conditions...).

* If the sample is large, the distribution of the least-squares estimators can be approximated by a normal, as both are linear combinations of the realizations of the dependent variable

Residuals

Regression residuals:

$$e_i = (y_i - \hat{y}_i) = (y_i - b_0 - b_1 x_i)$$

The residuals analysis can be conducted based on the original residuals or on the residuals:

- ▶ **standardised** using the standard error of residuals

$$\frac{e_i}{s_\varepsilon} = \frac{(y_i - \hat{y}_i)}{s_\varepsilon}$$

- ▶ **standardised** using a residual-specific standard deviation estimate for each residual

$$s_{e_i} = s_\varepsilon \sqrt{1 - h_i} \quad \text{with} \quad h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{(n-1)s_x^2}$$

where h_i , the so-called leverage, measures the deviation of x_i from the mean $\bar{x}$ based on the relative contribution of x_i to the variance s_x^2

$$\frac{e_i}{s_{e_i}} = \frac{(y_i - \hat{y}_i)}{s_{e_i}} = \frac{(y_i - \hat{y}_i)}{s_\varepsilon \sqrt{1 - h_i}}$$

Rstudio: analysis of residuals

From the **output of the `lm()`** function, it is possible to obtain some **plots of the residuals** using:

```
plot(object, which)
```

where **object** is the output of the function `lm()` and **which** indicates the type of graph to be obtained. We limit attention to some of these plots (see later for others):

which=1 scatterplot: relationship between residuals and predictions of the dependent variable

which=2 q-q plot of residuals. Plot of the quantiles of the standardised residuals against the quantiles of the normal distribution.

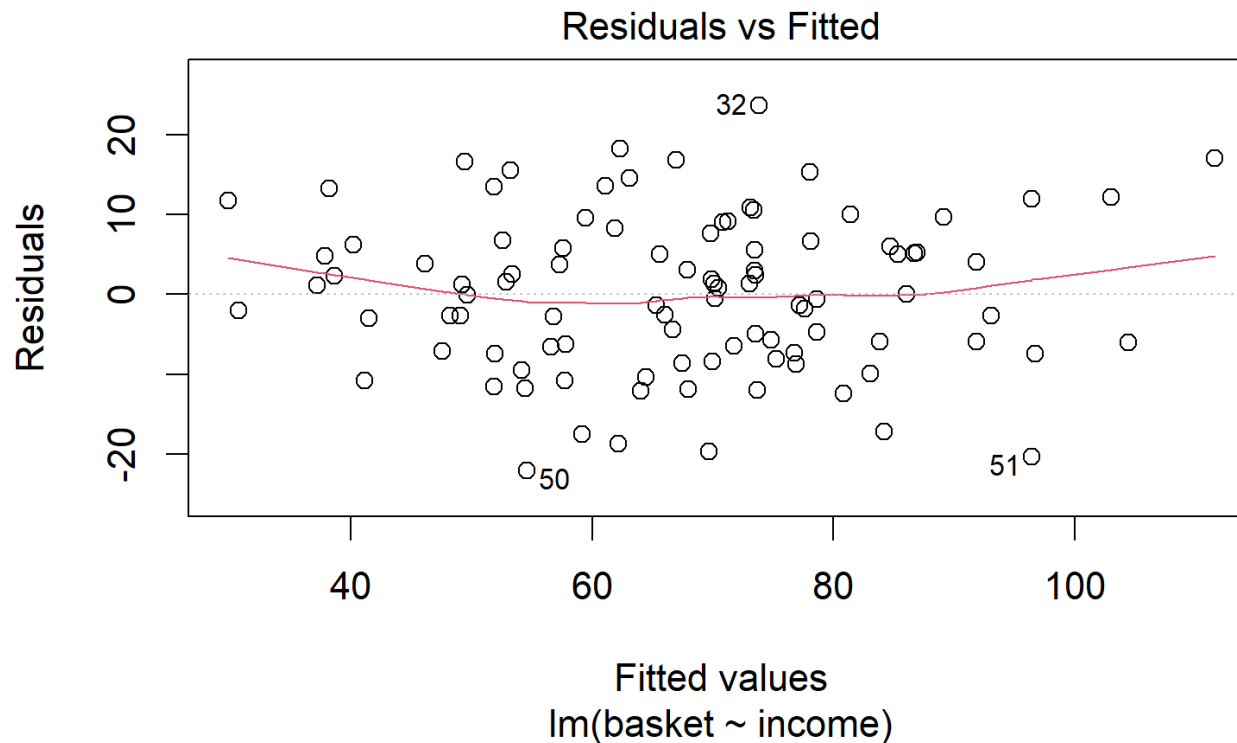
which=3 scale-location plot. Plot of the square root of the standardized residuals (in absolute value) against the predictions of the dependent variable

Note that the output of the function `lm()` also contains the regression residuals (`object$residuals`). The standardised residuals can be obtained using the function `rstandard(object)`

Residuals: structural characteristics

Relationship between **income** and quality of expenditure (**basket**).

```
> plot(sreg.model, which=1)
```

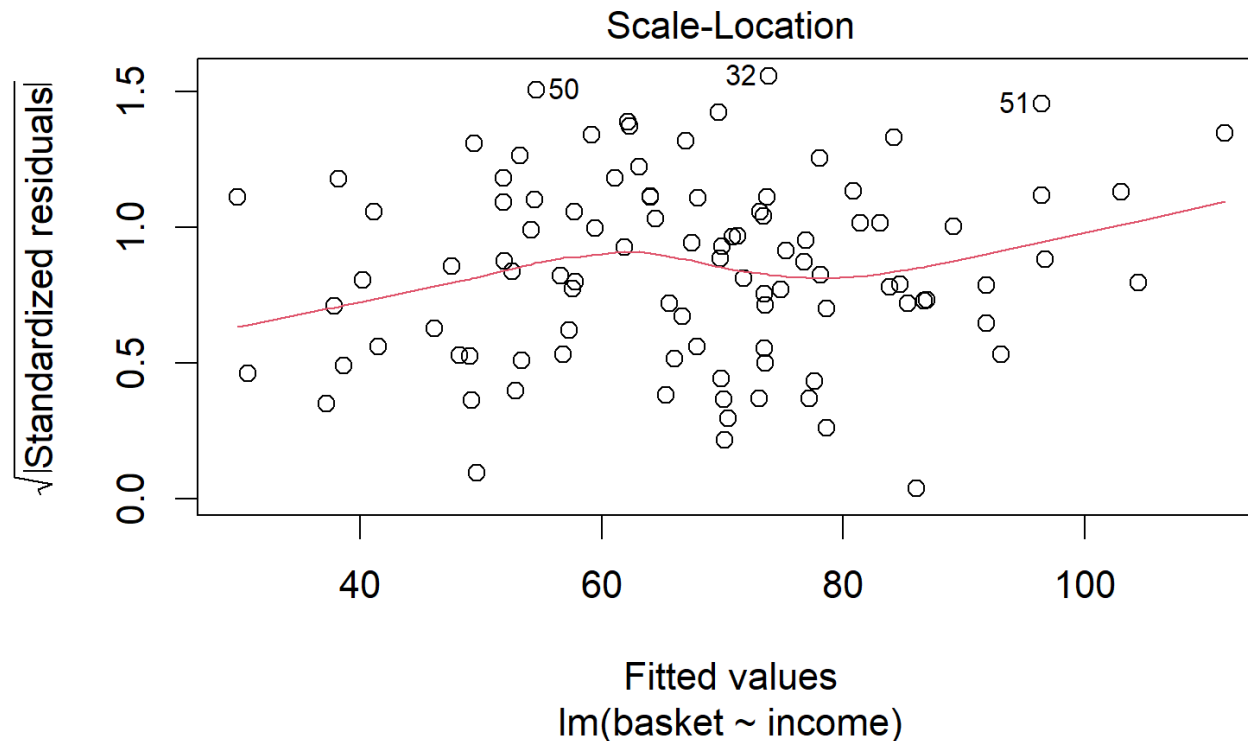


The residuals should *have no structure*.
The mean of the residuals is **zero** by construction; it must be assessed whether the mean level of the residuals changes at certain expected values. Furthermore, the variance of the residuals should be constant. We do not observe any particular structure of residuals, although some cases, highlighted in the plot, have relatively large residuals and corresponding to higher and lower predictions the (local) average of residuals is slightly higher than in the centre

Residuals: structural characteristics

Relationship between **income** and quality of expenditure (**basket**).

```
> plot(sreg.model, which=3)
```

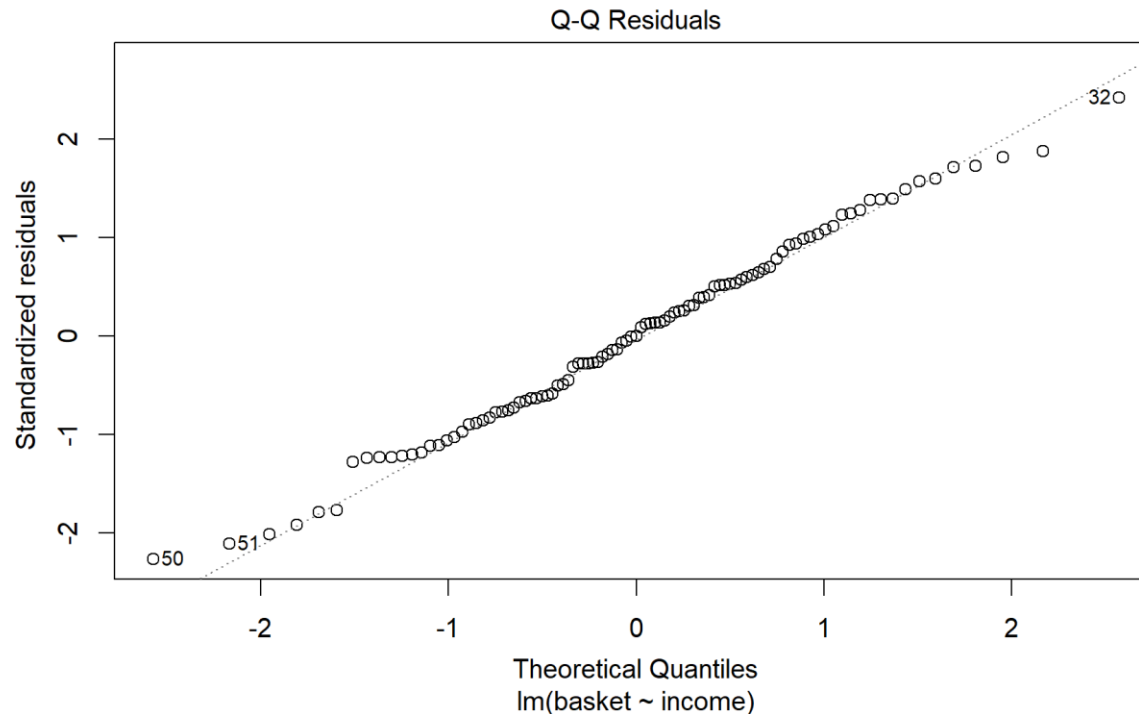


The residuals should *have no structure*. The magnitude of (absolute) residuals and their dispersion should not vary with predictions. Also in this case no particular structure can be noted. The average magnitude of (absolute) residuals is quite constant, even if slightly lower and higher corresponding respectively to lower and higher predictions. A slight reduction of the dispersion corresponding to higher predictions can be also appreciated.

Residuals: distribution - comparison with normal distribution

Relationship between **income** and quality of expenditure (**basket**).

```
> plot(sreg.model, which=2)
```



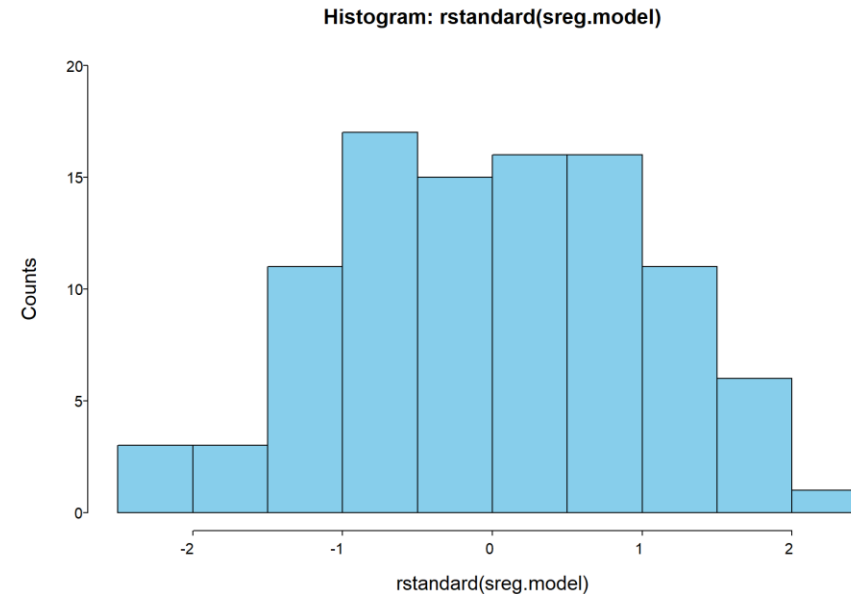
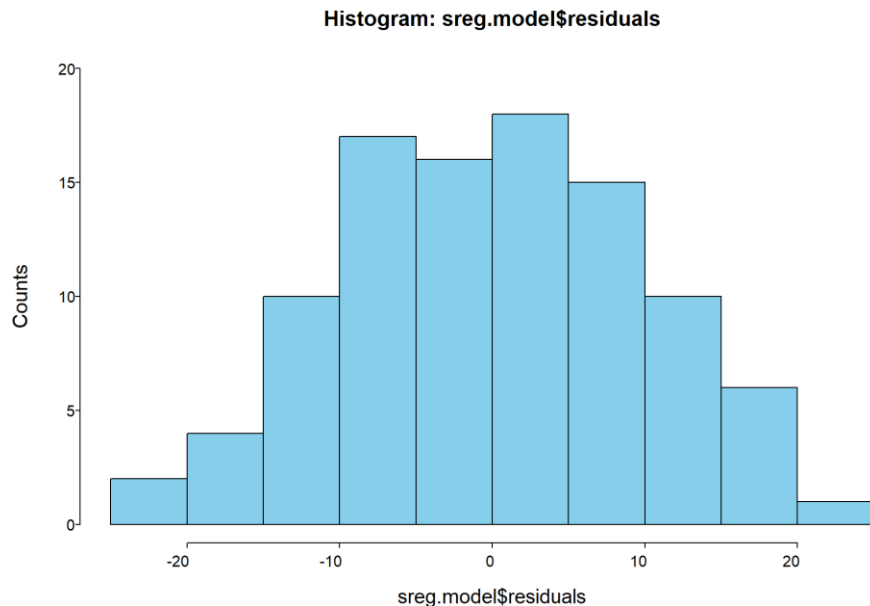
Although not strictly necessary for inferential purposes as the sample size is sufficiently large, we check whether the residuals have an approximately normal distribution, particularly to assess skewness and possible tails.

The distribution is approximately normal, although some deviation is observed on the tails. The left tail is slightly 'heavier' (quantiles are higher than those of the normal distribution) and the right tail is slightly 'lighter'.

Residuals: distribution

Relationship between **income** and quality of expenditure (**basket**).

```
> distr.plot.x(sreg.model$residuals,plot.type="histogram")  
> distr.plot.x(rstandard(sreg.model),plot.type="histogram")
```

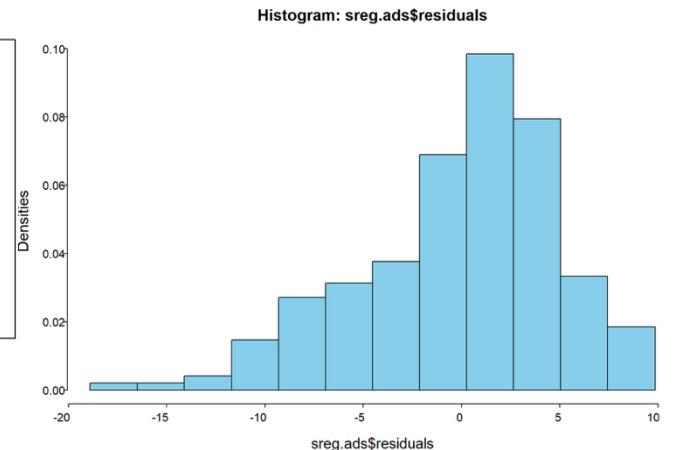
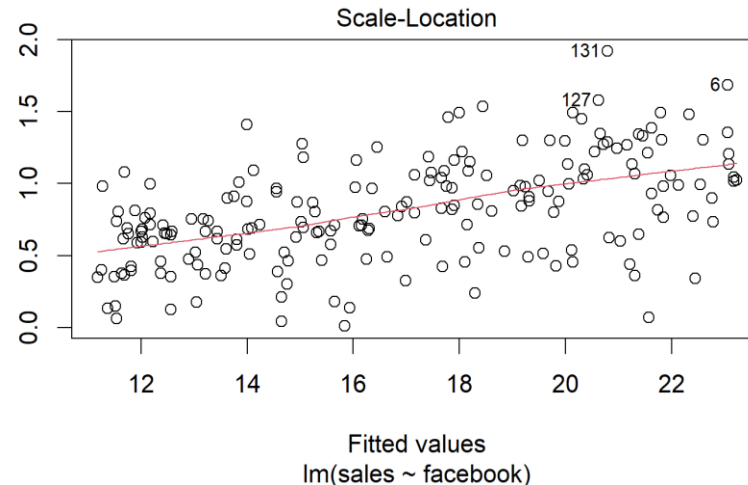
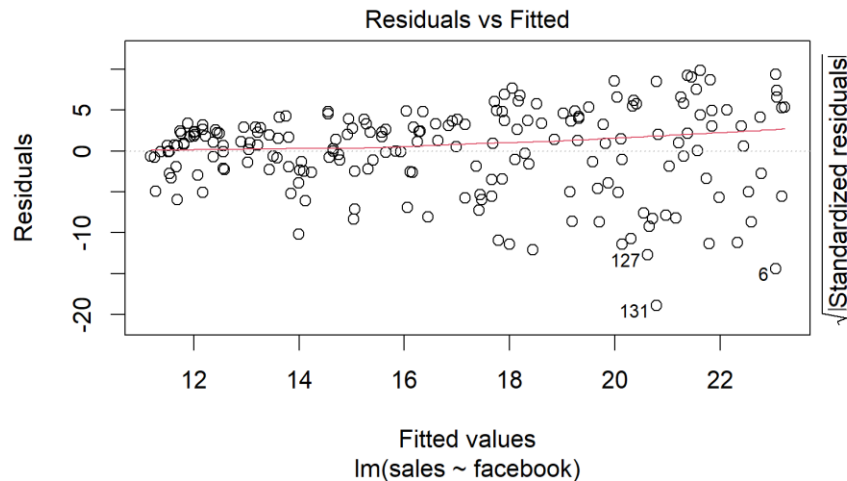


A simpler graphical analysis of the distribution of residuals, possibly standardised, is based on histograms. One can easily appreciate that in this case the distribution is quite symmetric even if the shape of the tails is not particularly 'normal'.

Analysis of residuals: some examples

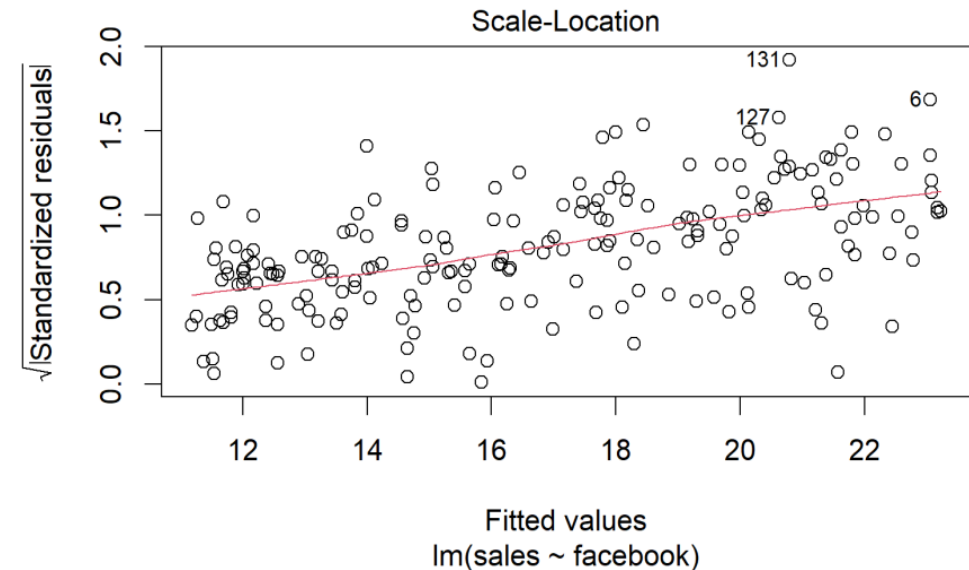
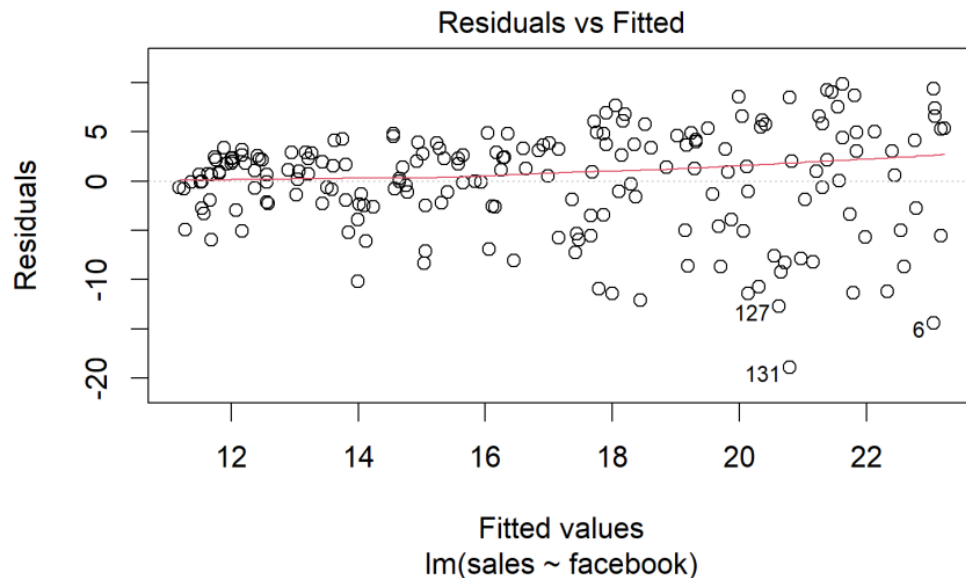
The dataframe **Advertising** contains information on the amount in thousands of dollars invested in advertising on different media (**youtube**, **facebook**, **newspapers**) and sales (**sales**, appropriately measured and indexed). Evaluate the assumptions underlying the linear model of **sales** on **facebook**

```
> sreg.ads<-lm(sales ~ facebook , data=Advertising)
> plot(sreg.ads,which=1)
> plot(sreg.ads,which=3)
> distr.plot.x(sreg.ads$residuals, plot.type="histogram", breaks=12)
```



Analysis of residuals: some examples

sales and advertising investment on **facebook**



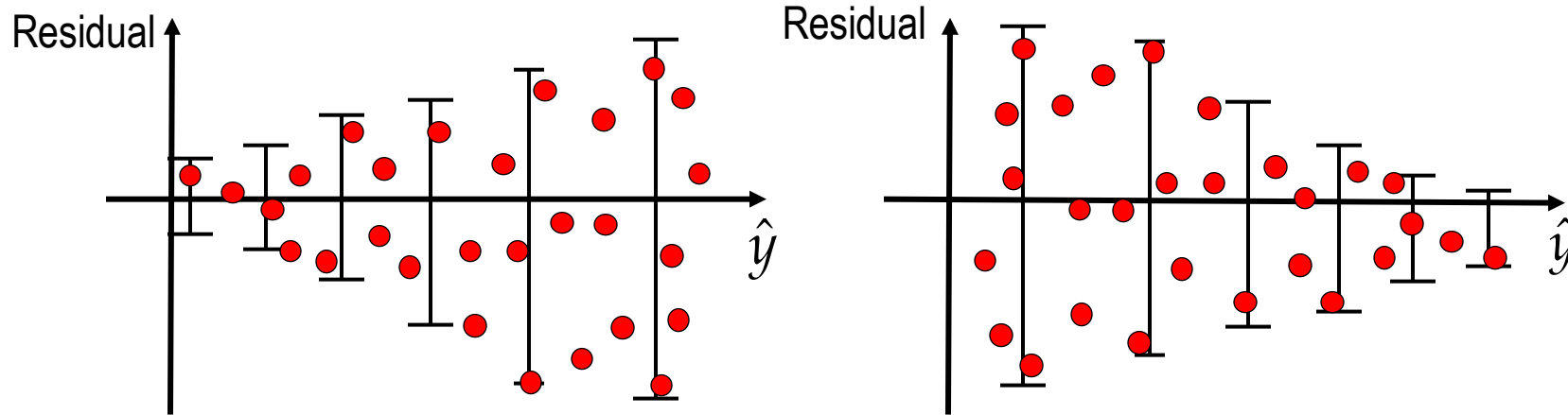
Observations?

We note an increase in the 'local' average of residuals and a greater dispersion of residuals around the average for higher values of the prediction. The increase in magnitude is really evident in the scale-location plot.

This indicates possible violation of the hypothesis of homoskedasticity (heteroskedasticity)

Assumption of homoskedasticity

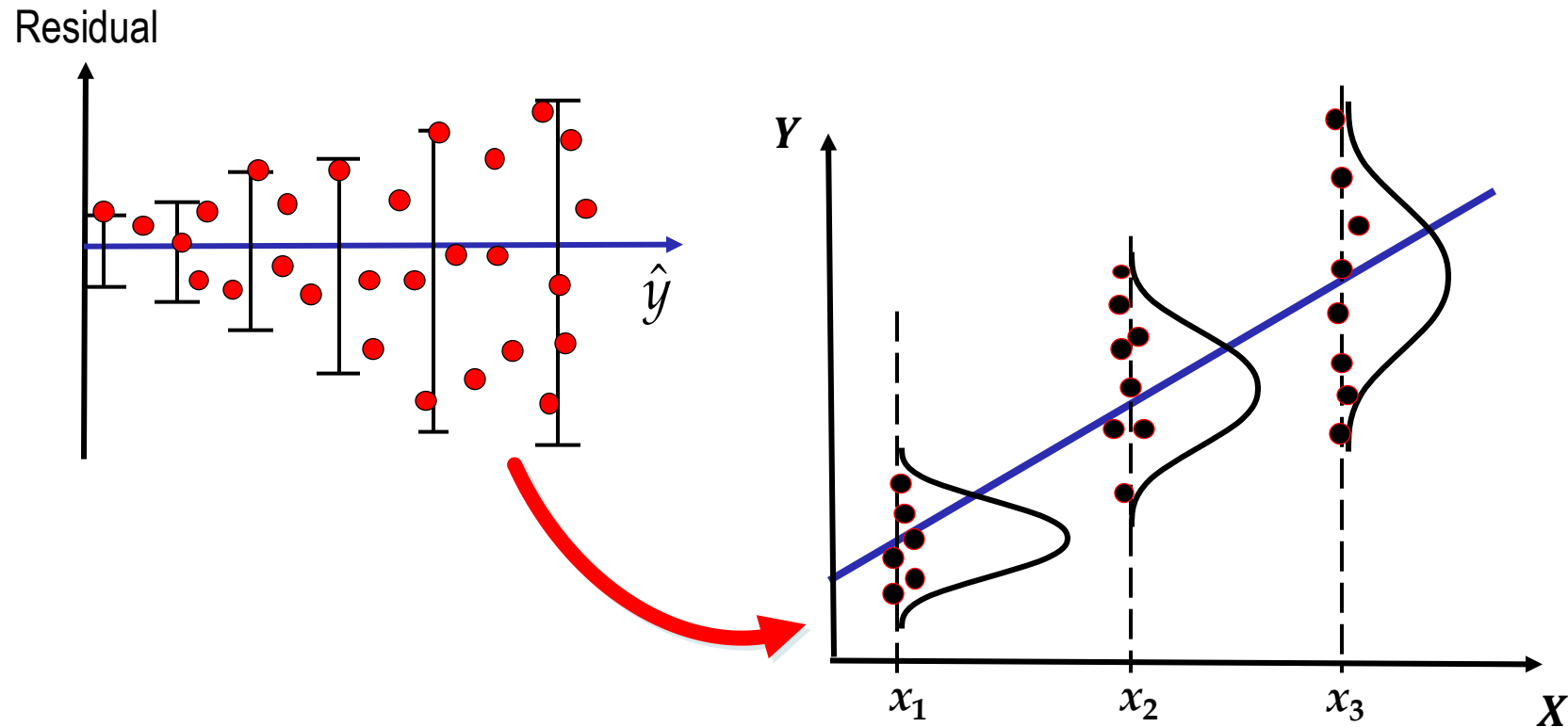
Plotting the residuals against the predicted values can signal situations where the assumption of homoskedasticity is violated.



The dispersion of the residuals varies as $\hat{y}$ varies (and therefore as x varies in the case of a simple linear model). In the case of heteroskedasticity, the scale-location plot will exhibit a patterned average of the square roots of absolute residuals and a variation in their dispersion.

Assumption of homoskedasticity

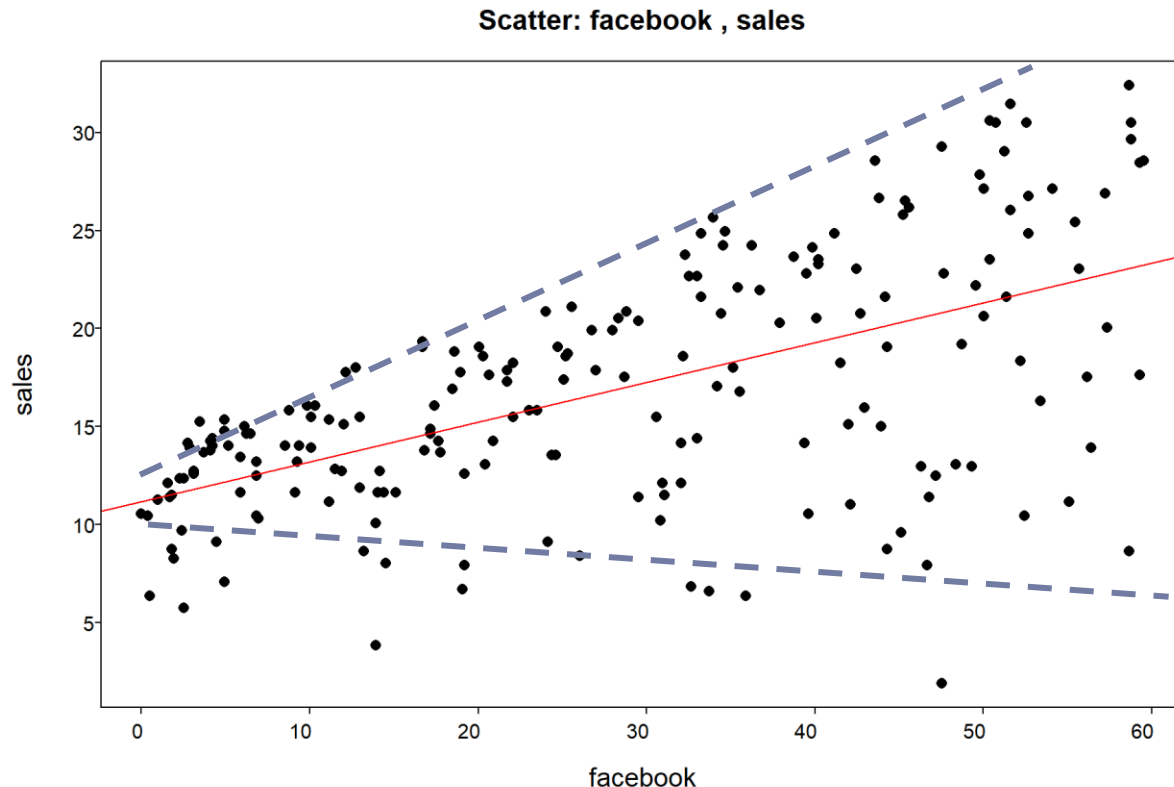
Violation of the assumption of homoskedasticity (heteroskedasticity): the variance varies when X varies



Assumption of homoskedasticity

sales and advertising investment on **facebook**

In the case of the simple linear model, linearity and homoschedasticity can be verified immediately from the data.

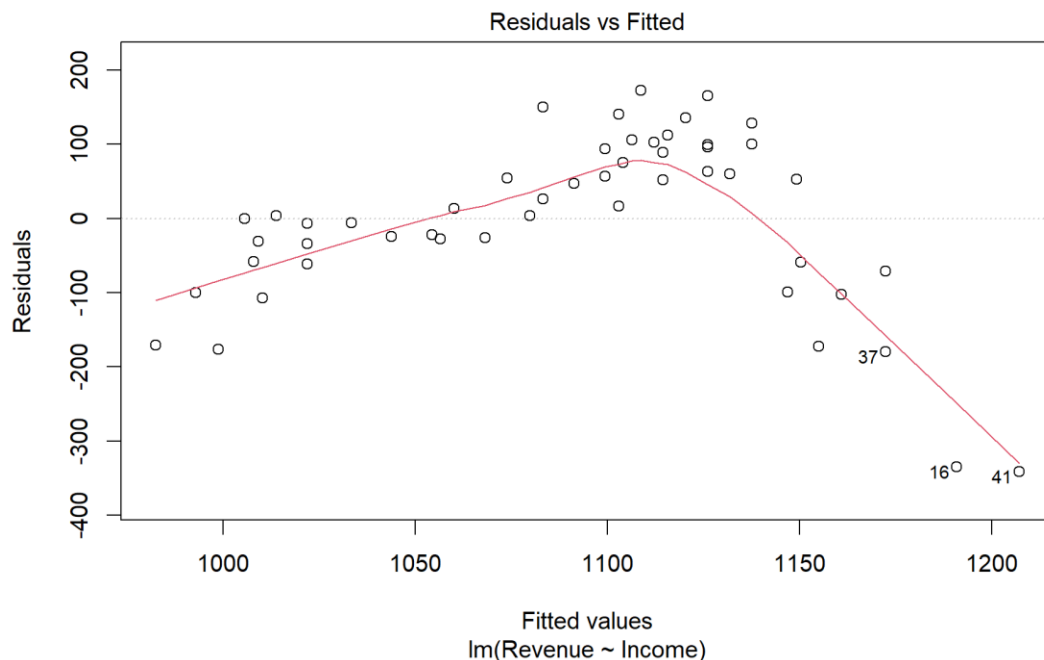


We present the analysis of residuals in this simplified context because when **several explanatory variables are** considered (multiple regression), the linear model cannot be represented graphically as simply and therefore the analysis of residuals can prove very useful.

Analysis of residuals: some examples

The dataframe **Restaurants** contains information on the revenues of the restaurants of a fast-food chain (**revenue**) and on the characteristics of the neighbourhood in which they are located, in particular the average **income**

Evaluate the assumptions underlying the linear model of **revenue** on income*.



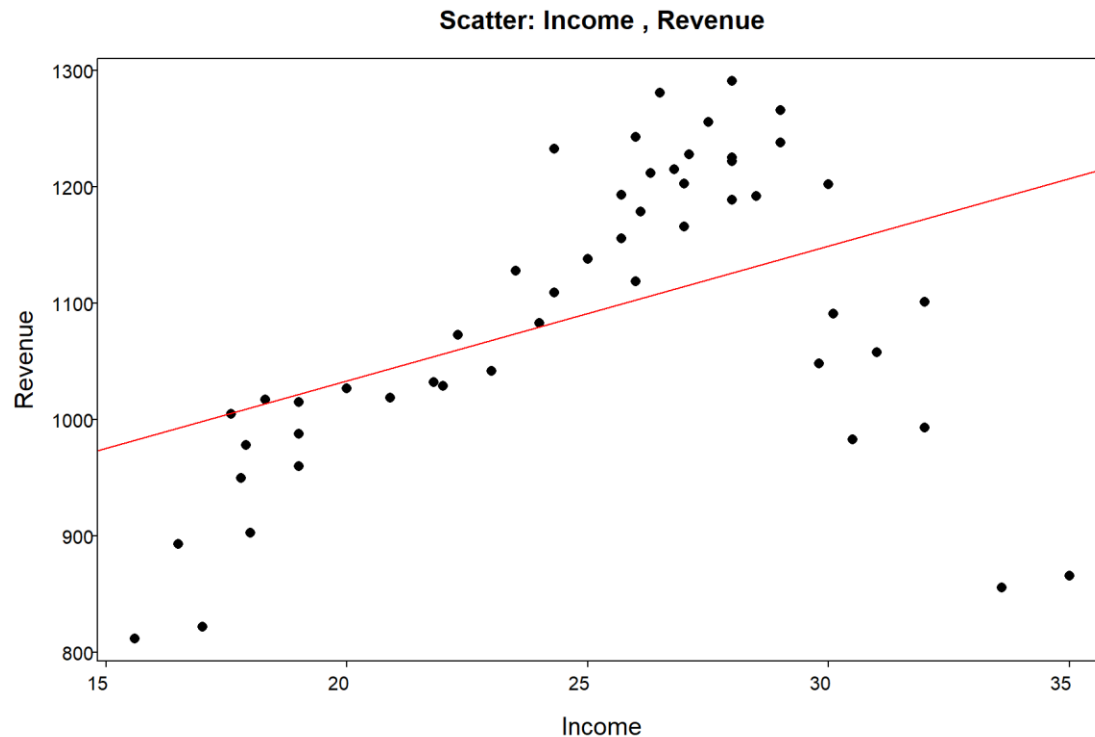
The residuals are characterised by an obvious non-linear structure: in particular, **revenue** is underestimated at lower values of **income** and heavily and systematically underestimated for higher values of **income**; this indicates that the observed values of **revenue** for high or low values of **income** deviate from the straight line!

Same conclusions can be drawn based on the scale-location plot* and the distribution of residuals is clearly not normal*

Analysis of residuals: non-linearity

Receipts (**revenue**) average **income** of the district in which they are located

Again, in order to deduce non-linearity, it was sufficient to consider the scatterplot of the two variables from the beginning, which is always recommendable



The plot clearly shows the non-linear structure of the relationship: from a certain point onwards **revenue** decreases as **income** increases because in 'richer' neighbourhoods the propensity to visit fast food restaurants is lower; the same is observed for 'less wealthy' neighbourhoods

Note: residuals analysis allows non-linear relationships to be identified even for models with several explanatory variables that cannot be represented graphically using a scatterplot

Simple linear regression: summary

- ▶ Develop a model that has a reasonable theoretical basis, and collect data on the variables involved
- ▶ **Descriptive analysis**
 - ▶ Analyse the scatter plot to assess whether the linearity assumption makes sense
 - ▶ Determine the regression line
 - ▶ Assess the presence of anomalous observations influencing the model to a too great extent
 - ▶ Assess the quality of the model

At this level, the analysis is purely **descriptive**.

To make **inference** about the model at the population level, certain assumptions must be made

Simple linear regression: summary

- ▶ Assess whether assumptions are violated by analysing residuals. There are clearly more sophisticated procedures to assess assumptions, as well as specific approaches developed to suitably modify the model in the event of a violation.
- ▶ If the assumptions are not violated, the estimated model can be used for inference
 - ▶ Check the significance of the coefficients
 - ▶ If one is interested in prediction, **it is important to consider that a significant coefficient does not imply that the model fits the data well (the R-squared should be used at this aim).**
 - ▶ Avoid extrapolation
 - ▶ Prefer confidence intervals to point estimates for forecasting
 - ▶ Choose the proper type of interval (confidence and prediction) based on the goal of analysis/prediction.